



*Project Report On*

**JobNexus: An AI-Powered Hiring and Career  
Assistance Platform**

*Submitted in partial fulfillment of the requirements for the  
award of the degree of*

**Bachelor of Technology**

*in*

***Computer Science and Engineering***

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# CERTIFICATE

*This is to certify that the project report entitled “**Job Nexus: An AI-Powered Hiring and Career Assistance Platform**” is a bonafide record of the work done by **Nandana Anukumar(U2203147)**, **Nikhitha Konnikara Chandran(U2203160)**, **Rinu Tony (U2203180)**, **Sruthilaya Sunil Kumar (U2203200)**, submitted to Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in “Computer Science and Engineering” during the academic year 2025-2026.*

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# Abstract

Recruitment platforms often rely on manual resume screening and keyword-based filtering, which can lead to inefficient hiring and poor alignment between candidate skills and job requirements. This project introduces JobNexus, an AI-powered recruitment and career assistance platform designed to improve hiring efficiency through intelligent automation and semantic analysis. JobNexus supports both candidates and recruiters. Candidates can search for jobs, build or upload resumes, view AI-based match scores, perform skill-gap analysis, receive career recommendations, and practice interviews using AI-generated questions with feedback. Recruiters can post and manage job listings, review applicants, rank candidates using semantic similarity scoring, and manage application statuses with automated notifications. The system's intelligence is powered by multiple AI modules.

Resume-job matching uses sentence embeddings from the all-MiniLM-L6-v2 model with cosine similarity to generate match scores. Skill-gap analysis identifies missing competencies and suggests courses using a FAISS-based engine, while career recommendations are generated using a machine learning pipeline based on TF-IDF and multinomial logistic regression. By integrating resume building, job matching, skill analysis, interview preparation, and career recommendations into a single platform, JobNexus enhances recruitment transparency and helps candidates make more informed career decisions.

Compared to conventional recruitment systems that depend heavily on keyword matching and manual evaluation, JobNexus introduces semantic analysis, AI-driven career guidance, and automated candidate evaluation within a unified platform. By integrating resume building, job matching, skill analysis, interview preparation, and career recommendations into a single ecosystem, the system improves recruitment transparency and supports informed job applications.



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## List of Abbreviations

|               |                                                                |
|---------------|----------------------------------------------------------------|
| <b>AI</b>     | – Artificial Intelligence                                      |
| <b>NLP</b>    | – Natural Language Processing                                  |
| <b>ATS</b>    | – Applicant Tracking System                                    |
| <b>BERT</b>   | – Bidirectional Encoder Representations from Transformers      |
| <b>ESCO</b>   | – European Skills, Competences, Qualifications and Occupations |
| <b>EURES</b>  | – European Employment Services                                 |
| <b>O*NET</b>  | – Occupational Information Network                             |
| <b>LLM</b>    | – Large Language Model                                         |
| <b>GAN</b>    | – Generative Adversarial Network                               |
| <b>LGIR</b>   | – LLM-based Generative Interactive Recommendation              |
| <b>KNN</b>    | – K-Nearest Neighbor                                           |
| <b>TF-IDF</b> | – Term Frequency–Inverse Document Frequency                    |
| <b>GNN</b>    | – Graph Neural Network                                         |
| <b>ERU</b>    | – Efficient Resume Understanding                               |
| <b>ESGNet</b> | – Entity Semantic Graph Network                                |
| <b>RAG</b>    | – Retrieval-Augmented Generation                               |
| <b>CV</b>     | – Curriculum Vitae                                             |
| <b>JD</b>     | – Job Description                                              |
| <b>nDCG</b>   | – Normalized Discounted Cumulative Gain                        |
| <b>RBO</b>    | – Ranked Biased Overlap                                        |
| <b>MAP</b>    | – Mean Average Precision                                       |
| <b>API</b>    | – Application Programming Interface                            |
| <b>GPU</b>    | – Graphics Processing Unit                                     |
| <b>LLaMA</b>  | – Large Language Model Meta AI                                 |
| <b>UML</b>    | – Unified Modeling Language                                    |
| <b>FAISS</b>  | – Facebook AI Similarity Search                                |
| <b>REST</b>   | – Representational State Transfer                              |

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# Chapter 1

## Introduction

The rapid growth of digital recruitment platforms has significantly transformed the way organizations search for and evaluate potential candidates. However, many existing recruitment systems still rely heavily on manual resume screening and simple keyword-based filtering methods. These approaches often fail to capture the true semantic meaning of candidate profiles and job descriptions, leading to inefficient hiring decisions and poor candidate-job alignment.

This project presents **JobNexus**, an AI-powered hiring and career assistance platform designed to improve the recruitment process through intelligent automation and data-driven insights. The system integrates semantic resume-job matching, skill-gap analysis, course recommendations, career recommendations, and interview preparation into a single platform. By leveraging machine learning and natural language processing techniques, JobNexus provides both candidates and recruiters with tools that enhance hiring efficiency and career development.

### 1.1 Background

Online recruitment platforms have become an essential part of modern hiring processes. Organizations frequently receive hundreds or even thousands of job applications for a single position, making manual screening time-consuming and inefficient. Traditional recruitment tools primarily depend on keyword matching and manual evaluation to shortlist candidates. While these approaches provide basic filtering, they often fail to understand the context and meaning of skills mentioned in resumes and job descriptions.

Recent advancements in artificial intelligence, natural language processing, and machine learning have opened new opportunities to enhance recruitment systems. Semantic text representations, such as sentence embeddings, allow systems to understand the con-

ceptual similarity between documents rather than relying only on exact keyword matches. These techniques can improve candidate evaluation, automate screening, and provide meaningful insights into skill alignment.

At the same time, candidates often struggle to understand how well their resumes match a job role or what skills they need to improve. Many recruitment platforms focus only on job discovery and application submission without providing personalized career guidance. Integrating AI-driven insights such as skill-gap detection, career recommendations, and interview preparation can significantly enhance the usefulness of recruitment platforms for both candidates and recruiters.

Therefore, there is a growing need for intelligent recruitment platforms that combine job discovery, resume analysis, and career development support. JobNexus aims to address these limitations by integrating AI-based semantic matching, machine learning models, and career assistance tools within a unified system.

## **1.2 Problem Definition**

Traditional recruitment systems rely on manual resume screening and simple keyword-based filtering, which often results in inefficient candidate evaluation and poor alignment between job requirements and candidate skills. The aim of this project is to develop an AI-powered recruitment platform that improves candidate-job matching and provides intelligent career guidance using semantic analysis and machine learning techniques.

## **1.3 Scope and Motivation**

The scope of this project is to design and develop an AI-powered recruitment and career assistance platform that supports both candidates and recruiters. The system enables candidates to search and apply for jobs, manage resumes, calculate AI-based match scores, perform skill-gap analysis, and receive career recommendations. Recruiters can post job listings, review applications, rank candidates using semantic similarity scoring, and manage hiring workflows. The platform integrates multiple AI modules such as resume-job semantic matching, course recommendation systems, and career prediction models to enhance the recruitment process.

The motivation for this project arises from the limitations of traditional recruitment



platforms that rely heavily on manual evaluation and keyword-based filtering. Candidates often lack clarity about how well their resumes match job requirements, while recruiters spend significant time reviewing applications manually. By integrating artificial intelligence techniques such as natural language processing, semantic similarity analysis, and machine learning, the proposed system aims to automate candidate evaluation and provide meaningful insights that support better hiring decisions and career development.

## **1.4 Objectives**

- To design and develop a multi-role recruitment platform that supports candidates and recruiters.
- To implement semantic resume–job matching using sentence embeddings and similarity scoring.
- To develop a skill-gap analysis module that identifies missing skills from candidate resumes.
- To provide course recommendations for improving missing skills.
- To implement a career recommendation system using machine learning techniques.
- To develop an AI-based interview preparation module that evaluates candidate responses.

## **1.5 Challenges**

One of the main challenges in the project is accurately extracting and interpreting skills from unstructured resume text and job descriptions. Another challenge involves designing a reliable semantic similarity scoring mechanism that produces meaningful match scores while handling variations in resume formats and wording.

## **1.6 Assumptions**

The system assumes that resumes contain sufficient textual information to extract relevant skills and experiences. It is also assumed that job descriptions provided by recruiters

clearly specify the required skills and responsibilities needed for accurate matching and analysis.

## **1.7 Societal / Industrial Relevance**

The proposed system has significant relevance for both industry and society. For organizations, it improves recruitment efficiency by automating candidate screening and ranking processes. For job seekers, the platform provides valuable insights such as skill-gap analysis, career recommendations, and interview preparation tools, helping them make informed career decisions and improve their employability.

## **1.8 Organization of the Report**

This report is organized into multiple chapters that describe the design, implementation, and evaluation of the proposed system. Chapter 1 introduces the project, including the background, problem definition, objectives, and motivation. Chapter 2 presents a literature survey of existing recruitment systems and related AI techniques. Chapter 3 describes the system design, architecture, modules, and technologies used. Chapter 4 presents the results and discussion of the system performance. Finally, Chapter 5 concludes the report and discusses future improvements and extensions.

## **1.9 Summary of Chapter**

This chapter introduced the concept of an AI-powered recruitment and career assistance platform called JobNexus. It discussed the background of modern recruitment systems, defined the problem addressed by the project, and described the objectives, scope, and motivation of the work. The chapter also highlighted the challenges, assumptions, and societal relevance of the system, providing a foundation for the subsequent chapters of the report.

## Chapter 2

### Literature Survey

#### 2.1 Introduction

Recent advancements in Artificial Intelligence and Natural Language Processing (NLP) have transformed automated recruitment. Traditional platforms often struggle with keyword limitations, failing to capture the semantic nuances between candidate profiles and job descriptions. This chapter reviews contemporary research—ranging from transformer-based embeddings and multimodal document understanding to Large Language Model (LLM) frameworks—that addresses these challenges in resume understanding and job recommendation.

#### 2.2 Related Works

##### 2.2.1 CareerBERT: Matching Resumes to ESCO Jobs in a Shared Embedding Space [1]

CareerBERT is a transformer-based model designed to bridge the gap between candidate qualifications and job roles using a shared semantic embedding space. By leveraging BERT architectures, the model moves beyond simple attribute matching to capture the contextual relationships between skills and experience. It specifically integrates the European Skills, Competences, Qualifications and Occupations (ESCO) taxonomy with real-world data from the EURES platform to ensure standardized yet market-relevant job recommendations.

#### Methodology

The CareerBERT workflow maps resumes and job descriptions into high-dimensional dense vectors. When a resume is uploaded, the text is extracted, preprocessed, and en-

coded via a BERT-based transformer. Simultaneously, job descriptions from ESCO and EURES are converted into vectors within the same embedding space. The system then calculates the **cosine similarity** between the resume vector and job vectors. Roles are ranked based on these scores, with higher values indicating a stronger semantic alignment between the candidate’s profile and the job requirements.

### Advantages

- **Semantic Understanding:** Captures context rather than just keyword overlaps.
- **Improved Matching Accuracy:** Shared embedding spaces allow for direct, effective similarity computation.
- **Structured Data Integration:** Utilizes the ESCO taxonomy for reliable, standardized occupational definitions.
- **Market Relevance:** Learns from actual EURES job postings to stay current with industry trends.

### Limitations

- **Regional Dependence:** Heavily tied to the European ESCO framework.
- **Domain Adaptation:** Requires significant retraining for non-European or niche industrial markets.
- **Sensitivity to Input:** Effectiveness relies on the structural quality of the uploaded resume.
- **Resource Intensive:** Transformer models require substantial computational power for inference.

#### 2.2.2 Resume2Vec: Intelligent Resume Embeddings for Precise Matching [2]

Resume2Vec is an AI framework developed to replace the keyword-centric logic of modern Applicant Tracking Systems (ATS). By utilizing various transformer encoders (BERT, RoBERTa, DistilBERT) and LLM decoders (GPT, LLaMA), it transforms unstructured

documents into dense semantic representations. The research highlights that these intelligent embeddings produce candidate rankings that align more closely with human recruiters' evaluations than traditional methods.

## Methodology

The framework utilizes a multi-stage pipeline: data collection, preprocessing, and embedding generation. The authors curated a dataset of 2,500 job descriptions via web scraping and paired them with public resume repositories. After cleaning the text (removing HTML, stop words, and non-ASCII characters), the system generates contextual vectors using both encoder-only and decoder-based models. Matching is performed via cosine similarity, and the system's performance is validated against human-ranked benchmarks using metrics like **nDCG** and **Ranked Biased Overlap (RBO)**.

## Advantages

- **Model Diversity:** Employs a range of transformers to capture deep semantic relationships.
- **Handling Unstructured Data:** Works across various resume formats without needing strict templates.
- **Human Alignment:** Experimental results show rankings closely mirror expert human judgement.
- **Bias Reduction:** Focuses on skill similarity rather than specific terminology.

## Limitations

- **Computational Costs:** High-end transformer models demand significant hardware resources.
- **Keyword Specificity:** In highly niche technical fields, traditional keyword matching may still occasionally outperform general embeddings.
- **Evaluation Constraints:** Initial qualitative tests relied on student evaluators rather than professional HR veterans.

### 2.2.3 Job Matching and Skill Recommendation Using O\*NET [3]

Alonso et al. present a framework that combines transformer-based text encoding with the structured knowledge of the Occupational Information Network (O\*NET) database. Beyond simple matching, this system focuses on "employability enhancement" by identifying skill gaps. By mapping resumes to the O\*NET knowledge graph, the system can suggest specific competencies a candidate lacks for a particular role.

#### Methodology

The system processes resumes and job descriptions through a transformer encoder to create shared-space embeddings. Crucially, it integrates a **Knowledge Graph** derived from O\*NET, which links occupations to required skills and abilities. By comparing a candidate's vector against the O\*NET-defined requirements for a specific role, the system uses cosine similarity for ranking and graph analysis to pinpoint missing skills, offering personalized development suggestions to the user.

#### Advantages

- **Skill Gap Identification:** Offers actionable insights by suggesting missing competencies.
- **Knowledge-Guided Matching:** Uses a formal occupational database to ground the AI's logic.
- **Contextual Accuracy:** Recognizes that different job titles often require identical skill sets.

#### Limitations

- **O\*NET Dependence:** The model's logic is limited by the scope and regional focus of the O\*NET database.
- **Integration Complexity:** Harder to implement within existing HR ecosystems due to the dual embedding-graph architecture.

- **Data Quality:** Performance is strictly bound by how well the user describes their initial skills.

#### 2.2.4 Enhanced Resume Analysis and Job Alignment Using BERT [4]

Swapna et al. propose a hybrid approach that merges the deep contextual power of BERT with the statistical reliability of Term Frequency–Inverse Document Frequency (TF-IDF). This dual-feature strategy aims to ensure that the system captures both the "meaning" of the text and the "importance" of specific technical keywords, which are often critical in specialized recruitment.

### Methodology

The architecture begins with a standard NLP preprocessing pipeline. It then generates two distinct feature sets: contextual embeddings from BERT and statistical weights from TF-IDF. These features are fused to represent the resume and job description. Similarity is calculated using cosine distance, and the **K-Nearest Neighbor (KNN)** algorithm is employed to classify the candidate into the most relevant job categories. The system was deployed via a Streamlit interface for real-time user interaction.

### Advantages

- **Hybrid Robustness:** Benefits from both semantic context and keyword weighting.
- **High Precision:** Reported a 92.3% precision rate in job alignment during testing.
- **Accessibility:** The Streamlit-based web interface makes the tool highly usable for non-technical users.

### Limitations

- **Scalability:** The KNN algorithm's efficiency decreases as the dataset size (number of resumes) grows.
- **Feature Engineering:** Balancing the weight between BERT and TF-IDF features requires careful tuning.

- **Industry Specifics:** May require domain-specific training to understand jargon in fields like medicine or law.

### 2.2.5 Job Recommendation through LLM-Based GANs [5]

The LGIR (LLM-based Generative Interactive Recommendation) framework tackles the problem of "information sparsity"—resumes that are too brief or incomplete. By combining LLMs with Generative Adversarial Networks (GANs), the system "fills in the blanks" by inferring missing candidate attributes based on the existing context, leading to a more robust profile for matching.

#### Methodology

The LGIR framework uses an LLM-powered **Interactive Resume Completion** module to infer likely skills or experiences missing from a sparse resume. These enriched profiles are converted into embeddings. A GAN architecture is then applied, where a **Generator** creates enhanced resume representations and a **Discriminator** attempts to distinguish these from high-quality, real candidate-job pairs. This adversarial process refines the embeddings, which are then ranked against job descriptions.

#### Advantages

- **Solves Data Sparsity:** Significantly improves recommendations for candidates with brief resumes.
- **Substantial Performance Gain:** Shows up to 35% improvement in ranking metrics for sparse data.
- **Enriched Representations:** Produces a more holistic view of a candidate's potential.

#### Limitations

- **Training Instability:** GANs are notoriously difficult to stabilize and prone to mode collapse.



- **Latency:** The dual process of LLM-based completion and GAN generation increases response times.

### 2.2.6 LLM Agents in Recruitment: A Multi-Agent Framework [6]

Gan et al. introduce a collaborative multi-agent system where specialized LLM "agents" take on different roles in the recruitment pipeline. This mimics a human HR team, with distinct agents handling parsing, evaluation, summarization, and ranking. This modularity allows for more transparent and structured decision-making.

#### Methodology

The framework assigns specific recruitment tasks to different LLM agents. A **Parsing Agent** extracts structured data; an **Evaluation Agent** compares skills to job requirements; a **Summarization Agent** creates brief candidate highlights; and a **Ranking Agent** assigns final scores. A central reasoning engine coordinates these agents, while reinforcement learning fine-tunes their behavior based on successful recruitment outcomes.

#### Advantages

- **End-to-End Automation:** Covers the entire pipeline from parsing to final ranking.
- **Modularity:** Agents can be updated or swapped without rebuilding the entire system.
- **Transparency:** The reasoning engine provides a more structured "thought process" for evaluations.

#### Limitations

- **Operational Overhead:** Running multiple agents simultaneously is computationally expensive.
- **Explainability:** While more structured, the underlying "black box" nature of LLMs remains a challenge.

### 2.2.7 ESGNet: Multimodal Network for Chinese Resume Extraction [7]

ESGNet addresses the complexity of resume layouts (tables, columns, and sections) by using a multimodal approach. Specifically designed for Chinese resumes, it integrates textual data with spatial layout features and semantic entity graphs to ensure that structural context—like where a job title is located—is not lost.

#### Methodology

The framework processes resumes through three channels: a BERT encoder for text, a layout analysis module for spatial coordinates, and an **Entity Semantic Graph**. This graph uses nodes to represent entities (e.g., "University," "Java") and edges for their relationships. **Graph Neural Networks (GNNs)** learn these relational patterns, and supervised contrastive learning is used to align these different modalities into a single, robust representation for information extraction.

#### Advantages

- **Structural Awareness:** Layout features prevent the loss of context found in document formatting.
- **High F1-Score:** Achieved 91.65% in extraction accuracy, outperforming text-only models.
- **Relational Logic:** Effectively links organizations to specific roles and dates.

#### Limitations

- **Data Intensive:** Requires large, layout-annotated datasets which are difficult to create.
- **Language Specificity:** Primarily optimized for Chinese; requires adaptation for other languages.

### 2.2.8 ERU: Multi-Granularity Multi-Modal Pre-Training Approach [8]

The Efficient Resume Understanding (ERU) framework focuses on the hierarchical nature of resumes. It uses multi-granularity learning to look at both the fine details (individual

words) and the global structure (sections and columns). This ensures that the system understands the document both as a sequence of text and as a visual object.

## Methodology

ERU uses a transformer-based encoder to process textual, visual, and structural data. It employs **cross-modal attention** to let the model "see" how a specific font size or location relates to the text content. The multi-granularity approach allows the model to process token-level details while maintaining an understanding of the document's overall hierarchical sections (e.g., identifying the "Experience" section before parsing the dates within it).

## Advantages

- **Hierarchical Depth:** Better at interpreting complex, non-linear resume layouts.
- **Competitive Performance:** Surpasses models like LayoutLMv3 in resume-specific tasks.

## Limitations

- **GPU Intensive:** Multimodal pre-training and cross-modal attention require heavy computational resources.

### 2.2.9 Tailored Resume Generation Using RAG [9]

Patra et al. propose using Retrieval-Augmented Generation (RAG) to help candidates customize their resumes for specific jobs. Instead of a general resume, this system retrieves a candidate's relevant historical data and uses an LLM to rewrite or highlight sections that specifically match a target job description, ensuring the content is both tailored and factually grounded.

## Methodology

The system uses a two-part architecture: a **Retrieval Module** and a **Generative Model (LLaMA-3)**. When a job description is provided, the retriever pulls the most relevant skills and experiences from the candidate's master profile. These are fed as context to the

LLM, which generates a tailored resume. This "grounding" in retrieved data prevents the LLM from hallucinating experiences the candidate does not actually have.

### Advantages

- **Dynamic Tailoring:** Automatically emphasizes the most relevant qualifications for any given job.
- **Factually Grounded:** RAG reduces the risk of AI-generated "hallucinations."

### Limitations

- **Master Profile Quality:** The output is only as good as the initial data provided by the user.
- **Over-Optimization:** Risks making a resume look "too perfect" for a single job at the cost of broader appeal.

#### 2.2.10 ProspectCV: LLM-Based Advanced CV–JD Evaluation Platform [10]

ProspectCV is a multimodal evaluation platform that treats resumes as images to capture layout and text simultaneously. Using the Gemini Pro Vision model, it provides a comprehensive "score" for how well a candidate fits a job, along with feedback on what skills they are missing.

### Methodology

The platform converts resumes into Base64 encoded images. These images, alongside the text of the job description, are fed into the **Gemini Pro Vision** LLM. The model analyzes the multimodal input to assess skill alignment, education, and experience. It then generates a compatibility score (0–100) and produces a detailed feedback report via a Streamlit dashboard.

### Advantages

- **Vision-Text Integration:** Processes the resume as a visual document, preserving layout context.

- **Actionable Feedback:** Tells candidates exactly why they didn’t match and how to improve.

## Limitations

- **API Costs:** Reliance on proprietary multimodal LLM APIs can be expensive at scale.
- **Privacy:** Sending resumes to external APIs for processing poses security and data-handling concerns.

## 2.3 Summary of Chapter

This chapter reviewed recent advancements in AI-based recruitment systems, highlighting the shift from keyword-based methods to semantic and intelligent matching approaches. Models such as CareerBERT [1] and Resume2Vec [2] demonstrate the effectiveness of embedding-based similarity, while hybrid techniques combining BERT and TF-IDF [4] improve domain-specific accuracy. Knowledge-driven systems using O\*NET [3] further enhance matching by enabling skill gap identification.

Recent developments also include LLM-based frameworks [5, 6], Retrieval-Augmented Generation [9], and multimodal models like ESGNet [7] and ERU [8], which improve resume understanding by incorporating contextual, structural, and visual features.

Despite these advancements, challenges such as high computational cost, scalability issues, and data dependency remain. Overall, the reviewed works provide a strong foundation for developing intelligent, scalable, and user-centric recruitment systems, influencing the design of the proposed JobNexus platform.

# Chapter 3

## System Design

This chapter presents the overall design and architecture of the JobNexus platform. It describes how different system components interact to support AI-based recruitment and career assistance functionalities. The chapter focuses on system architecture, component design, and high-level workflow representation.

### 3.1 Overall Architecture

The JobNexus platform is built upon a modular, multi-stage pipeline that integrates Natural Language Processing (NLP), semantic vector analysis, and Large Language Models (LLMs) for specialized evaluation. The architecture is designed to transform unstructured career data into actionable insights through a six-stage processing flow, as illustrated in the system diagram.

#### 3.1.1 Architectural Stages

The system follows a sequence of data transformation, moving from raw text acquisition to high-level cognitive tasks such as interview evaluation and career path prediction.

#### Stage 1: Data Acquisition and Extraction

The pipeline begins with the **Data Acquisition** stage. Resumes and Job Descriptions (JDs) are fed into the system via the web interface. For PDF documents, the **pdfplumber** library is utilized as the core extraction engine due to its ability to handle complex multi-column layouts and tables, converting binary documents into raw, machine-readable text.

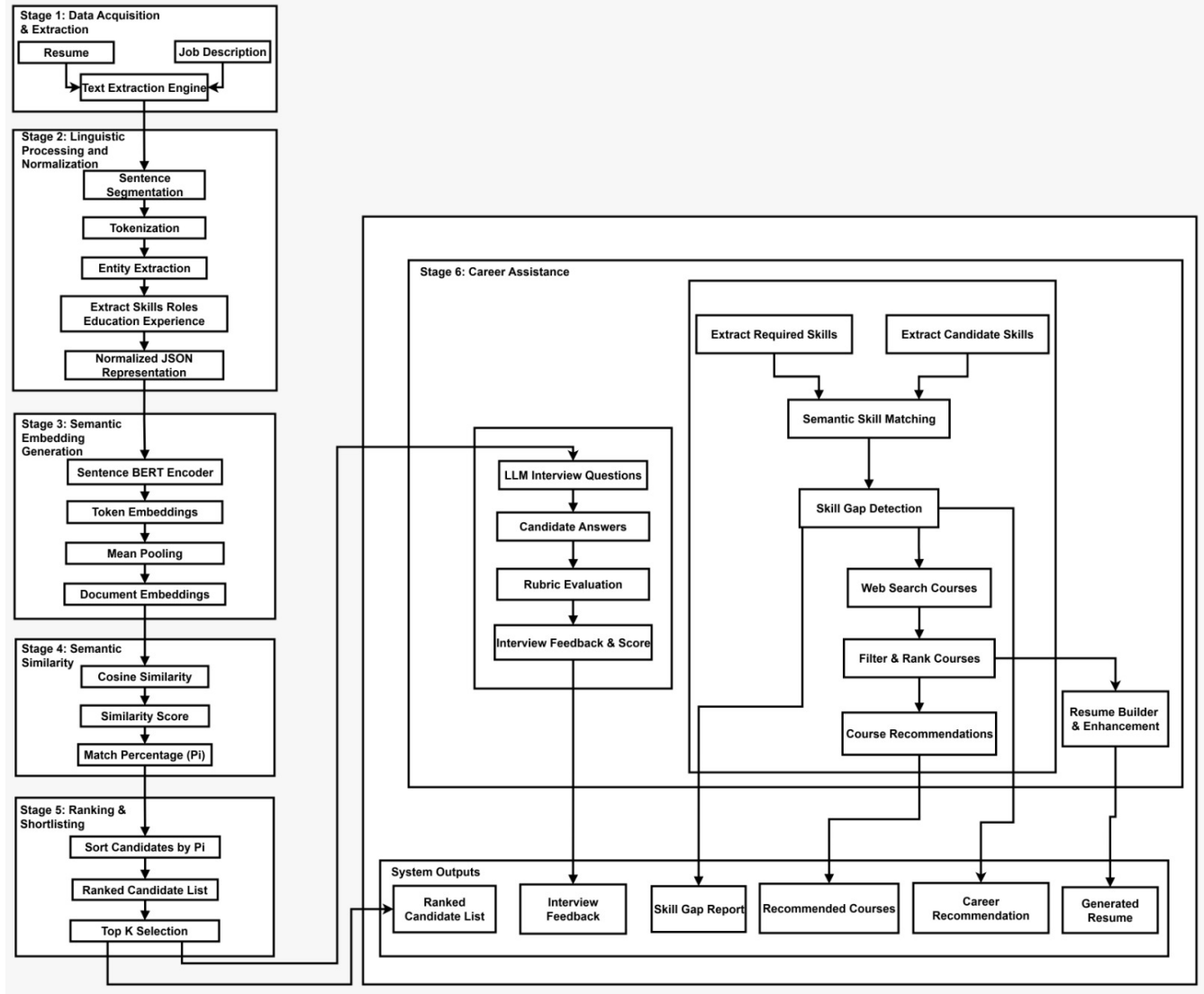


Figure 3.1: System Architecture of JobNexus

## Stage 2: Linguistic Processing and Normalization

Instead of generic parsing, the system employs a dedicated **Linguistic Processing Module** using **spaCy** and **NLTK**. This stage involves:

- **Preprocessing:** Tokenization, stop-word removal, and sentence segmentation.
- **Entity Extraction:** Identifying specific data points such as technical skills, job roles, and educational qualifications using Named Entity Recognition (NER).
- **Data Structuring:** The extracted linguistic features are organized into a **Normalized JSON Representation** to ensure consistency across the platform.

### Stage 3 & 4: Semantic Embedding and Similarity

The core of the JobNexus matching engine relies on dense vector representations:

- **Embedding Generation:** Using the **all-MiniLM-L6-v2** Sentence-BERT model, structured text is transformed into high-dimensional **Document Embeddings** that capture the contextual meaning of the profile.
- **Similarity Computation:** The system calculates the **Cosine Similarity** between the resume and JD vectors. This is processed at multiple levels (document-level, section-level, and skill-level) to produce a weighted **Match Percentage (II)**.

### Stage 5: Ranking and Selection

Once match percentages are calculated, the **Ranking & Shortlisting** layer sorts candidates in descending order. A **Top-K Selection** algorithm identifies the most relevant applicants, providing recruiters with a streamlined list based on semantic merit rather than simple keyword counts.

### Stage 6: Career Assistance and System Outputs

The final stage provides comprehensive support for candidates through three specialized sub-modules:

- **Interview Agent:** This module utilizes the **Gemini AI** model via a Flask-based microservice to generate role-specific interview questions and evaluate candidate responses for clarity and relevance.
- **Skill Gap & Learning Path:** By comparing extracted skill sets, the system identifies missing competencies. It uses **FAISS** (Facebook AI Similarity Search) to retrieve the most relevant resources from a course dataset.
- **Career Recommendation:** A **Logistic Regression** model trained on TF-IDF features predicts suitable career paths and probability scores for various professional roles.



The final **System Outputs** include the Ranked Candidate List, Interview Feedback, Skill Gap Reports, Recommended Courses, and Career Path Predictions. The complete system architecture is illustrated in Figure 3.1.

## 3.2 Design diagrams

The system design is represented using UML diagrams to visualize interactions, workflows, and system behavior.

### 3.2.1 Use Case Diagram:

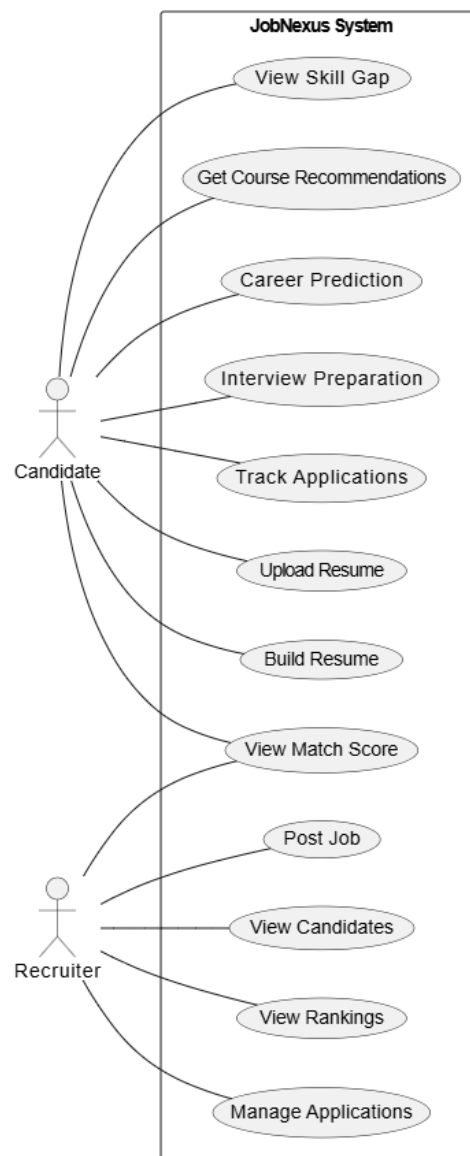


Figure 3.2: Use Case Diagram of JobNexus

The system consists of two primary actors: Candidate and Recruiter. Candidates can upload or build resumes, view match scores, identify skill gaps, receive recommendations, and track applications. This actor interacts with the system to optimize their professional profile through AI-driven insights. Recruiters can post jobs, view applicants, analyze rankings, and manage hiring processes. The diagram captures the functional requirements of the system, illustrating the boundaries between user actions and backend automated services, such as the matching engine and notification system. The interactions between actors and the system are shown in Figure 3.2.

### 3.2.2 Sequence Diagram:

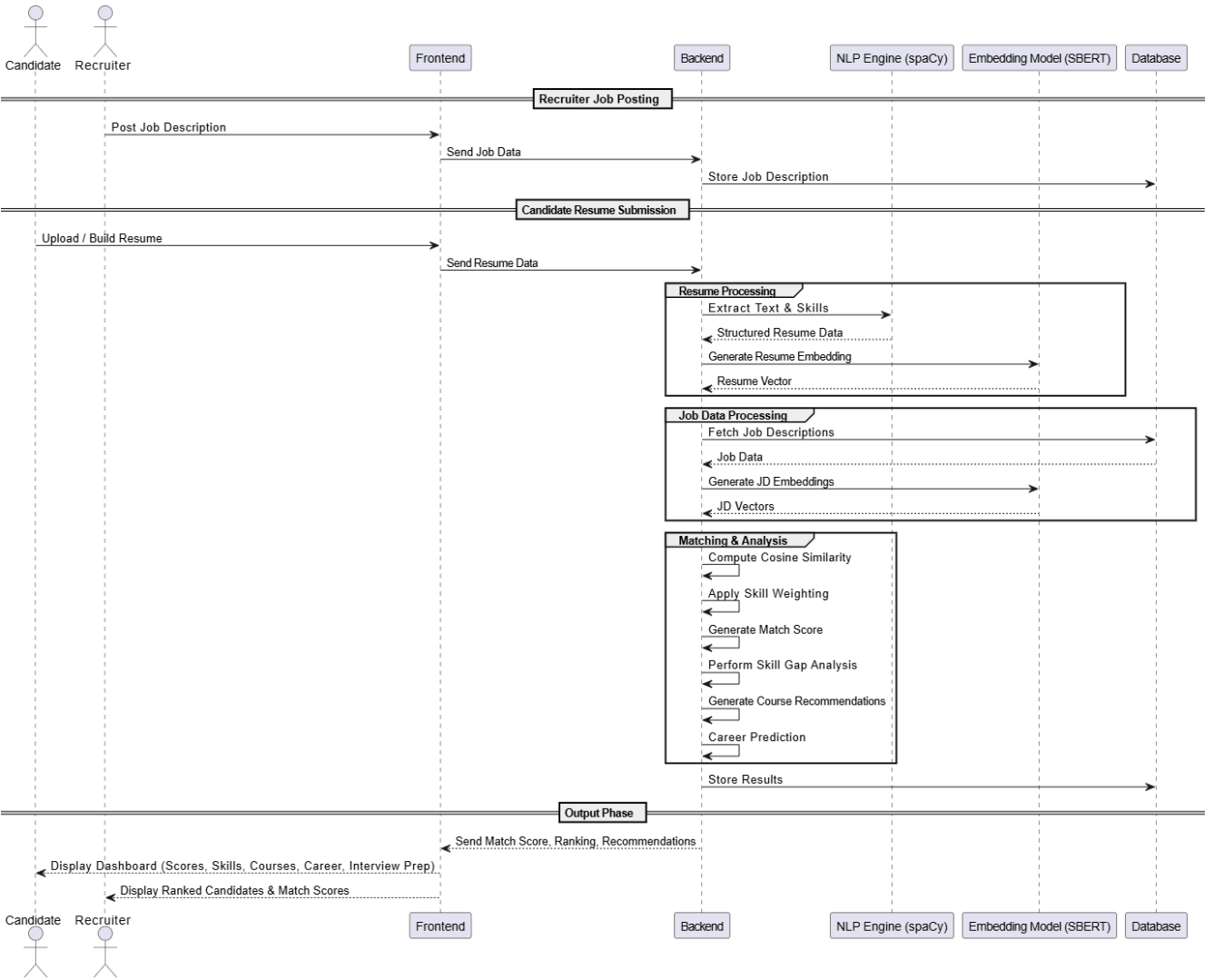


Figure 3.3: Sequence Diagram for Resume Matching

This diagram illustrates the interaction between components during resume processing and job matching. The process starts with resume upload, followed by parsing, embedding generation, similarity computation, ranking, and result display. It details the chronological flow of messages between the Frontend UI, the FastAPI backend, and the vector database. Specifically, it highlights how a candidate's document is converted into a high-dimensional vector, compared against job description embeddings using cosine Similarity, and how the resulting score is returned to the user interface in real-time. The sequence of operations is depicted in Figure 3.3.

### 3.2.3 Activity Diagram:

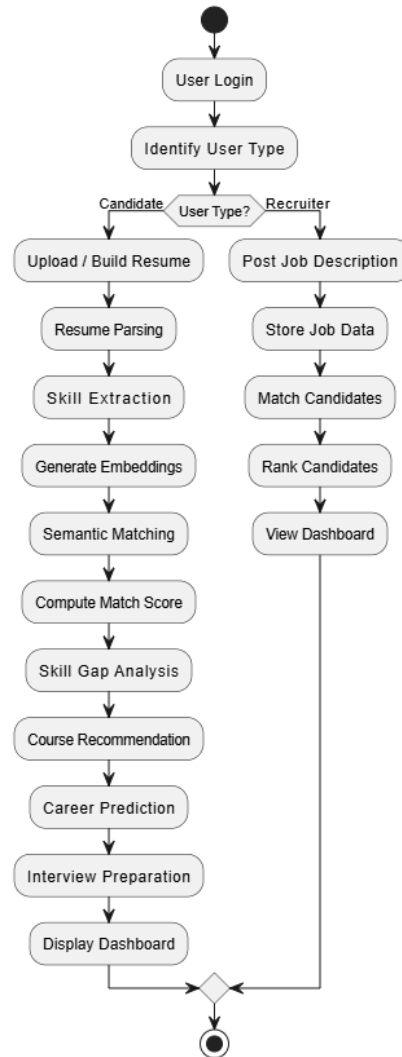


Figure 3.4: Activity Diagram of System Workflow

The activity diagram represents the workflow for both candidates and recruiters.

It includes authentication, resume processing, semantic matching, recommendation generation, and recruiter-side ranking and evaluation. The workflow maps the decision-making logic of the platform, such as the branching paths for a user who chooses to build a resume manually versus uploading an existing file. It also depicts the background synchronization of data, where the system continuously evaluates candidate profiles against new job postings to provide updated match notifications. The overall workflow is illustrated in Figure 3.4.

### **3.3 Summary of Chapter**

The JobNexus system is designed using a scalable client-server architecture with modular AI integration. The design clearly separates frontend, backend, and intelligent processing components, enabling efficient communication and maintainability. The inclusion of multiple functional modules ensures that key operations such as resume analysis, semantic matching, ranking, and recommendation are handled in a structured and organized manner.

UML diagrams, including use case, sequence, and activity diagrams, provide a comprehensive visualization of system interactions, workflows, and responsibilities. These representations help in understanding the dynamic behavior of the system as well as the flow of data between components.

Overall, the system design emphasizes modularity, scalability, and extensibility, allowing future enhancements such as improved AI models, additional recommendation features, and integration with external platforms. This structured design approach ensures clarity in system functionality and forms a strong foundation for the implementation phase.

# Chapter 4

## Implementation

This chapter presents a detailed description of the implementation of the JobNexus platform. It explains how each module is developed, the internal working of system components, and the technologies used to build a scalable AI-powered recruitment system. The implementation follows a modular approach where each component performs a specific function while interacting seamlessly with other modules through the backend.

### 4.1 Implementation of system modules

The JobNexus platform is implemented using a modular architecture in which each major functionality is designed as an independent module. This modular design improves system scalability, maintainability, and ease of integration. Each module processes specific data and passes it to subsequent modules through well-defined interfaces.

#### 4.1.1 Input and Data Collection Module

The Input and Data Collection module is responsible for acquiring data from both candidates and recruiters.

Candidates can either upload resumes in PDF format or create resumes using the built-in Resume Builder. The Resume Builder ensures structured input by collecting information through predefined sections such as personal details, education, experience, projects, and skills.

Recruiters provide job-related information including job title, job description, required skills, and additional preferences. All collected data is validated and stored securely in the MySQL database.

This module ensures consistency in data formatting and acts as the entry point for all downstream processing tasks.

#### 4.1.2 Data Parsing and Extraction Module

Since resumes are often uploaded as PDF documents, they must be converted into machine-readable text before processing.

Multiple libraries such as PyPDF, PyMuPDF, and pdfplumber were evaluated for text extraction. Among these, pdfplumber was selected due to its ability to handle complex layouts such as multi-column formats, tables, and varied font structures commonly found in resumes.

The extracted content is cleaned and preprocessed by removing noise such as extra spaces, special characters, and formatting inconsistencies. The output is converted into structured textual data, which is then forwarded to the NLP module for further analysis. Figure 4.1 illustrates the data processing and extraction workflow.

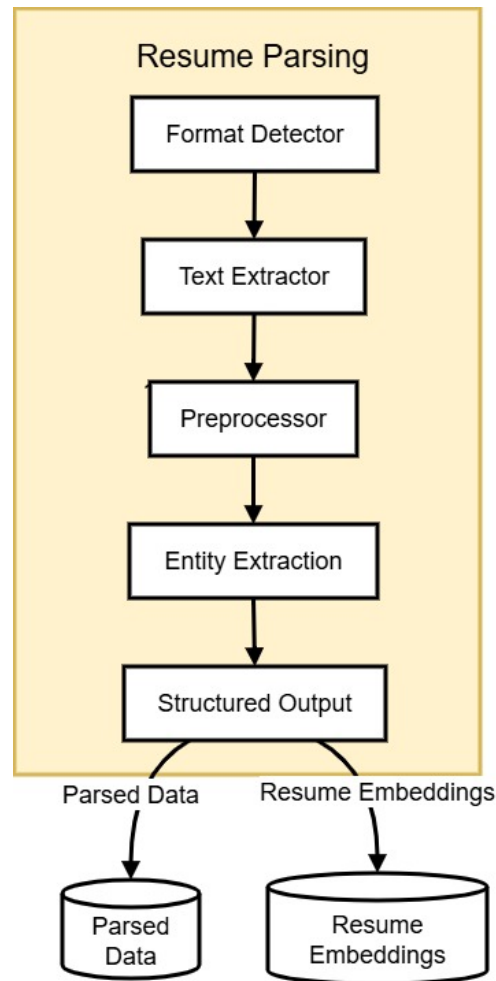


Figure 4.1: Data Processing and Extraction

### 4.1.3 Linguistic Processing Module

The Linguistic Processing module applies Natural Language Processing (NLP) techniques to extract meaningful information from resumes and job descriptions.

The implementation uses:

- **spaCy:** Used as the primary NLP engine for part-of-speech tagging, named entity recognition, phrase detection, and skill extraction.
- **NLTK:** Used for tokenization, stop-word removal, and sentence segmentation.

During processing, the system identifies key entities such as skills, technologies, job roles, and qualifications. It also extracts contextual phrases that improve semantic understanding.

This module transforms raw text into structured linguistic features that are used in downstream AI models. The linguistic workflow is depicted in Figure 4.2.

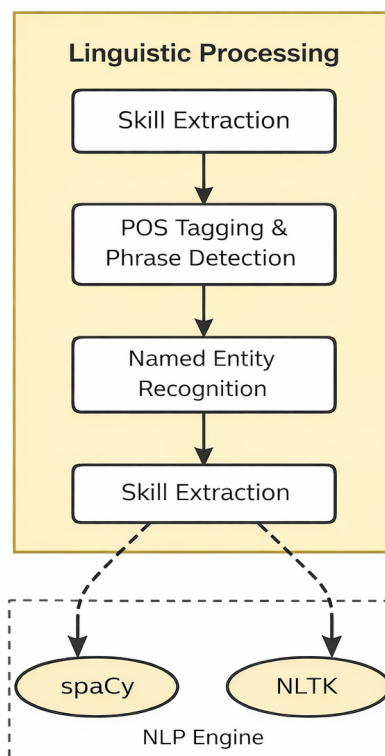


Figure 4.2: Linguistic Processing

#### 4.1.4 Resume Builder Module

The Resume Builder module provides an alternative to PDF uploads by enabling users to create structured resumes directly within the platform.

The module is implemented using a form-based interface where users input details across multiple sections. The data is stored in MySQL in JSON format, preserving the hierarchical structure of the resume.

Before processing, the JSON data is converted into standardized plain text. This normalization step ensures that all resumes—whether uploaded or generated—follow a consistent format, improving the accuracy of semantic matching. The interface of this module is shown in Figure 4.3.

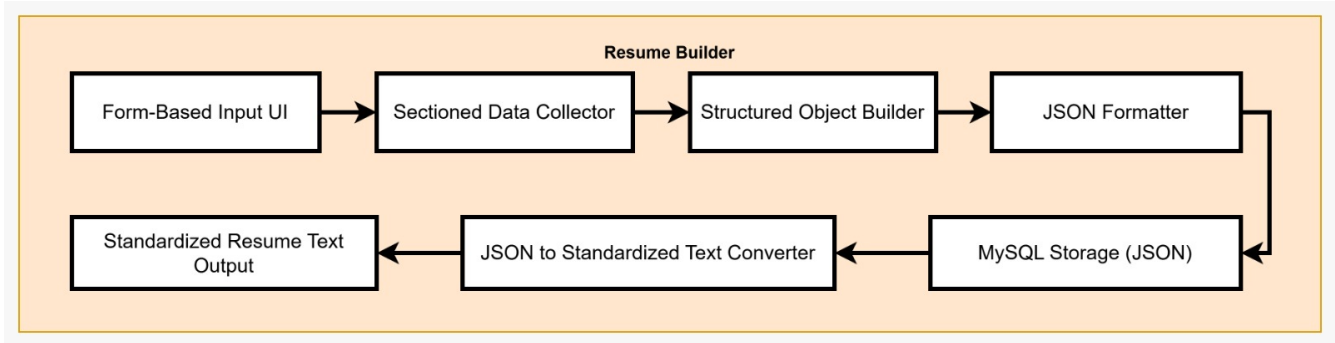


Figure 4.3: Resume Builder Module

#### 4.1.5 Semantic Resume–Job Matching Module

The Semantic Matching module is the core component of the system, responsible for evaluating how well a candidate’s profile matches a job description.

The implementation uses the Sentence Transformers library with the pretrained *all-MiniLM-L6-v2* model to generate dense vector embeddings for both resumes and job descriptions.

Instead of relying on a single similarity score, a hybrid scoring approach is implemented:

- **Document-level similarity:** Measures overall similarity between the entire resume and job description.
- **Section-level similarity:** Compares individual resume sections (skills, experience,



projects) with the job description.

- **Skill-level similarity:** Evaluates similarity between extracted skill sets.

Cosine similarity is computed for each level, and the results are combined using weighted scoring. Additional heuristics such as required skill boosting and missing skill penalties are applied to improve ranking accuracy.

The final score is normalized to a value between 0 and 1 and presented as a match percentage to the user. The overall semantic matching workflow is illustrated in Figure 4.4.

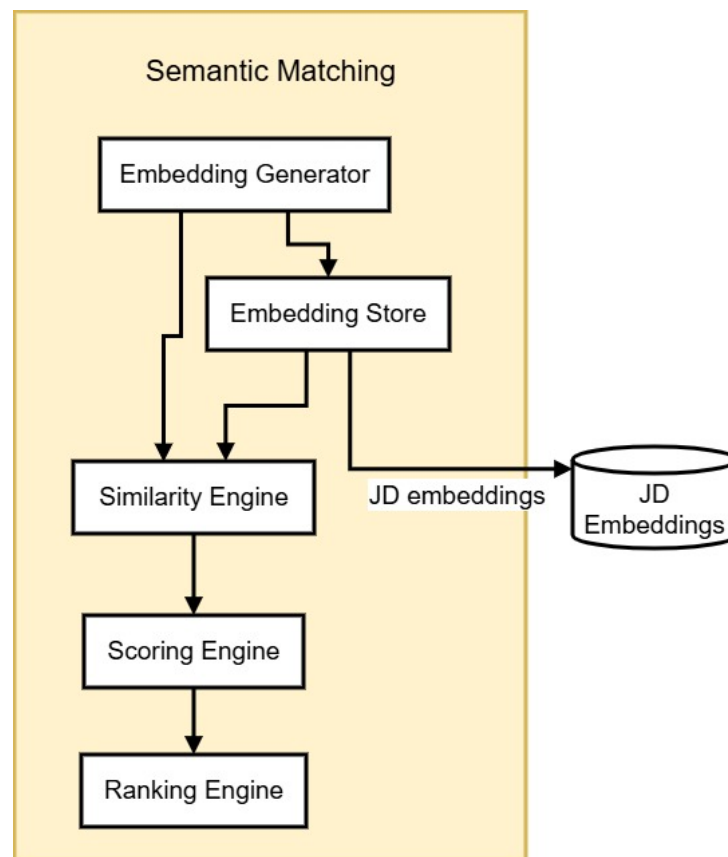


Figure 4.4: Semantic Matching Process

#### 4.1.6 Skill Gap Analysis Module

The Skill Gap Analysis module identifies the difference between required job skills and the candidate's existing skills.

The implementation uses a combination of:

- Exact keyword matching

- Lexical similarity techniques
- Embedding-based semantic similarity

Each required skill in the job description is evaluated against the candidate's skill set. Missing skills are identified and categorized into priority levels such as required, preferred, or optional.

The module computes a skill coverage score and generates actionable insights, helping candidates understand areas that require improvement. This process is visualized in Figure 4.5.

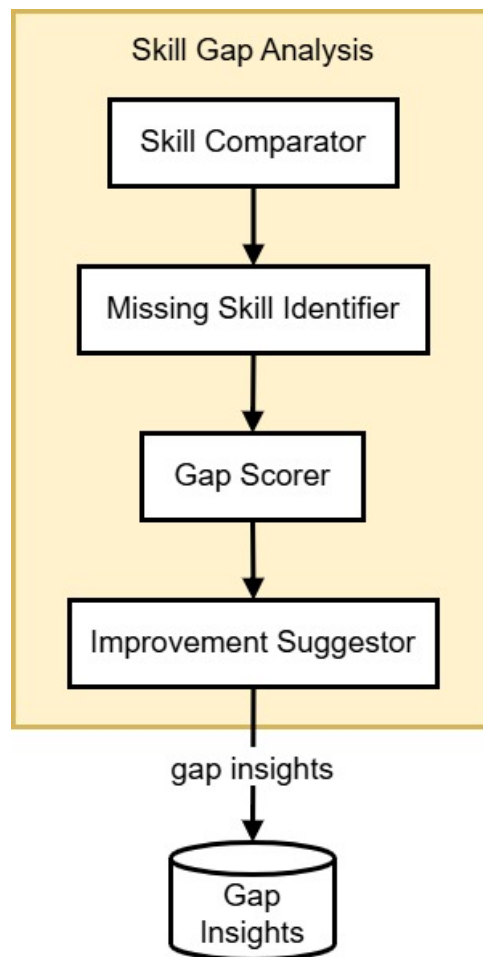


Figure 4.5: Skill Gap Analysis

#### 4.1.7 Course Recommendation Module

The Course Recommendation module suggests relevant learning resources based on identified skill gaps.

A local course dataset is maintained and converted into vector embeddings. These embeddings are indexed using FAISS (Facebook AI Similarity Search) for efficient nearest-neighbor retrieval.

When a missing skill is detected, the system generates an embedding for the skill query and retrieves the most relevant courses using vector similarity. The results are further refined using keyword-based ranking.

Additionally, the system can incorporate external resources from platforms such as Coursera, edX, and Class Central to enhance recommendation diversity. The recommendation workflow is shown in Figure 4.6.

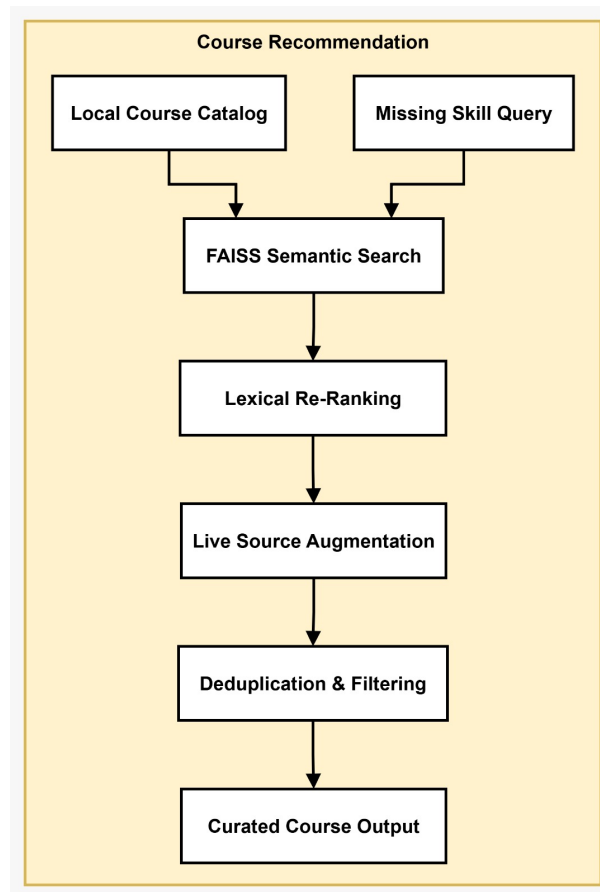


Figure 4.6: Course Recommendation System

#### 4.1.8 Interview Preparation Module

The Interview Preparation module enables candidates to practice interview questions tailored to specific job roles.

The system follows a session-based architecture:

- FastAPI manages session creation, storage, and lifecycle.
- A Flask-based microservice interacts with the Gemini AI model to generate questions and evaluate responses.

Each answer is analyzed and scored based on relevance, clarity, and completeness. The system provides structured feedback highlighting strengths and areas for improvement.

At the end of the session, an overall score is calculated as the average of individual question scores. The working of this module is illustrated in Figure 4.7.

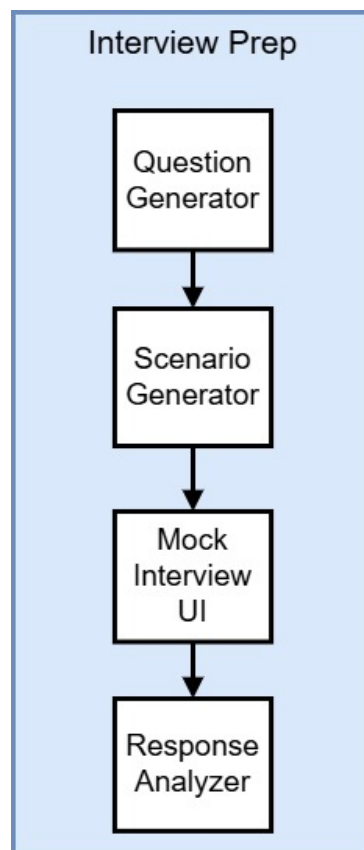


Figure 4.7: Interview Preparation Module

#### 4.1.9 Career Recommendation Module

The Career Recommendation module predicts suitable career paths based on resume content.

The implementation uses TF-IDF vectorization with unigrams and bigrams to convert text into numerical features. These features are processed using a multinomial Logistic Regression model trained on a curated dataset.

The model outputs probability scores for multiple career roles. The system presents the top recommendations along with alternative career paths and highlights the skills required for each role. This process is depicted in Figure 4.8.

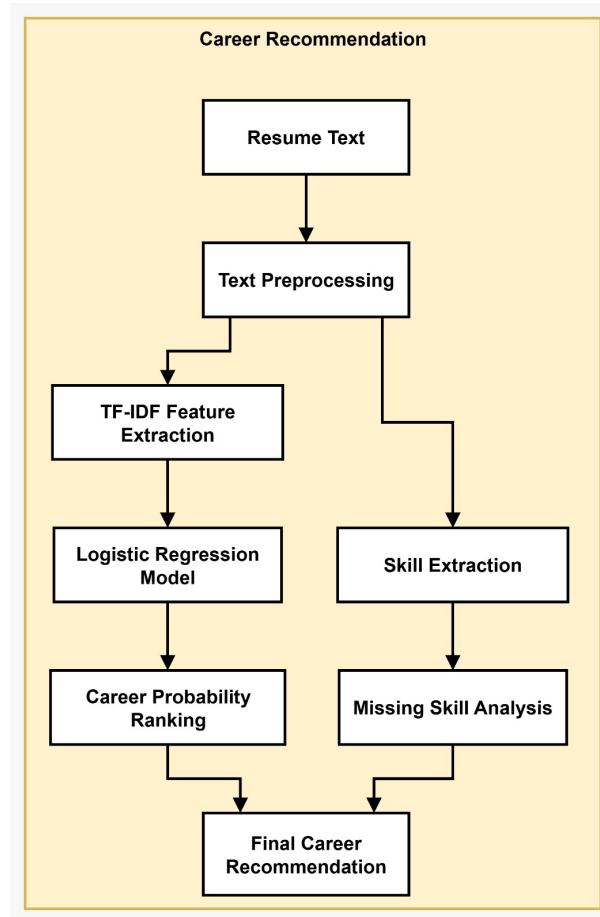


Figure 4.8: Career Recommendation System

#### 4.1.10 Candidate Ranking Module

The Candidate Ranking module is implemented as a scoring orchestration layer built on top of the semantic matching engine.

Each candidate receives a match score based on similarity with the job description. Candidates are sorted in descending order of scores to generate a ranked list.

Since the ranking logic directly uses outputs from the semantic matching module, consistency is maintained across evaluation and ranking processes.

## 4.2 Tools and Technologies

The JobNexus platform integrates multiple technologies across different layers:

- **Frontend Technologies:** React.js, HTML5, and CSS3 are used to build responsive and interactive user interfaces.
- **Backend Framework:** Python with FastAPI is used to implement high-performance RESTful APIs and handle communication between modules.
- **Database:** MySQL is used for storing user data, resumes, job postings, and system outputs.
- **Machine Learning Libraries:** Scikit-learn is used for classification models, while NumPy and Pandas support numerical computation and data manipulation.
- **Natural Language Processing Tools:** spaCy and NLTK are used for text processing and feature extraction.
- **Embedding Models:** Sentence Transformers (all-MiniLM-L6-v2) is used for semantic representation.
- **Vector Search:** FAISS is used for efficient similarity search in high-dimensional vector space.

## 4.3 Work Breakdown and Responsibilities

The project development was carried out collaboratively with clear distribution of responsibilities as seen in Figure 4.9:

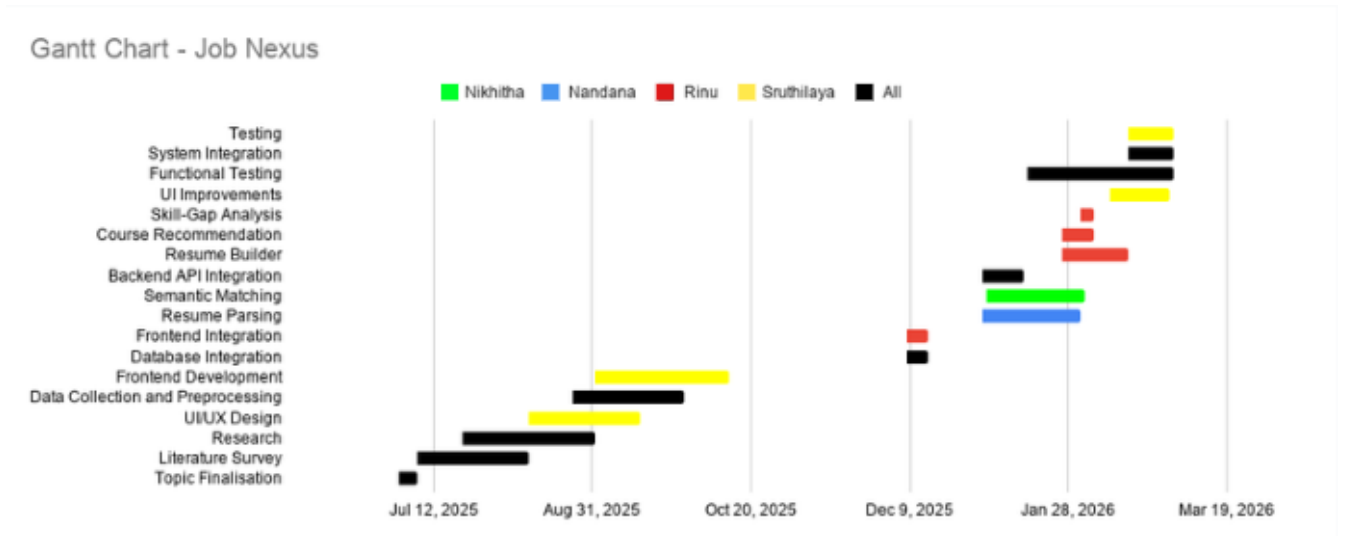


Figure 4.9: Gantt Chart

- **Nandana Anukumar:** Semantic matching, ranking logic, and career recommendation model
- **Nikhitha Konnikkara Chandran:** Resume parsing, NLP processing, and system integration
- **Rinu Tony:** Resume Builder, interview preparation module, skill gap analysis, course recommendation, and database management
- **Sruthilaya Sunil Kumar:** Frontend development, UI design, and testing

#### 4.4 Key Deliverables

The key deliverables of the project are as follows:

- Development of an AI-powered recruitment platform
- Implementation of a Resume Builder and resume management system
- Semantic resume-job matching with candidate ranking functionality
- Skill gap analysis integrated with a course recommendation system
- Career recommendation module for role prediction
- Interview preparation and automated evaluation system

- Comprehensive project documentation and final report

## 4.5 Summary of Chapter

This chapter provided a comprehensive overview of the implementation of the JobNexus platform, detailing how individual modules were designed, developed, and integrated to form a cohesive system. The implementation follows a modular architecture, where each component performs a well-defined function while communicating seamlessly with other modules through a FastAPI-based backend.

Key functionalities such as resume parsing, linguistic processing, semantic matching, ranking, and recommendation were implemented using a combination of Natural Language Processing (NLP) techniques, transformer-based embedding models, and machine learning algorithms. The integration of vector similarity search using FAISS further enhances the system's ability to efficiently retrieve relevant results from high-dimensional data.

The platform not only facilitates intelligent candidate-job matching but also extends its capabilities to provide career guidance through skill gap analysis, personalized course recommendations, and interview preparation support. This holistic approach ensures that the system serves both recruiters and candidates effectively.

Overall, the implementation demonstrates a robust, scalable, and efficient system design. The use of modern technologies and a well-structured architecture ensures high performance, maintainability, and flexibility, making JobNexus suitable for real-world deployment as well as future enhancements and feature expansions.



## Chapter 5

### Results and Discussions

This chapter presents the experimental results obtained from the implementation of the JobNexus platform and discusses the effectiveness of the proposed system. The platform was evaluated based on the performance of its machine learning components and the quality of its intelligent features such as resume classification, semantic job matching, skill gap analysis, and course recommendation. The results demonstrate how artificial intelligence and natural language processing techniques improve the recruitment process by enabling more accurate candidate-job matching and providing useful career guidance.

#### 5.1 Resume Classification and Career Recommendation

One of the key components of the JobNexus platform is the Career Recommendation module, which automatically analyzes a candidate's resume and predicts the most suitable professional domain as seen in Figure 5.1. This module uses machine learning algorithms to classify resumes into predefined job categories such as software development, data science, or business analysis.

Before classification, the textual content of resumes is preprocessed and converted into numerical feature vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique[11]. TF-IDF helps identify the importance of terms within a document relative to the entire dataset, thereby improving the performance of classification models.

Multiple machine learning models were evaluated in order to determine the most suitable algorithm for resume classification[4, 2]. The models tested include Naive Bayes, Random Forest, and Logistic Regression.

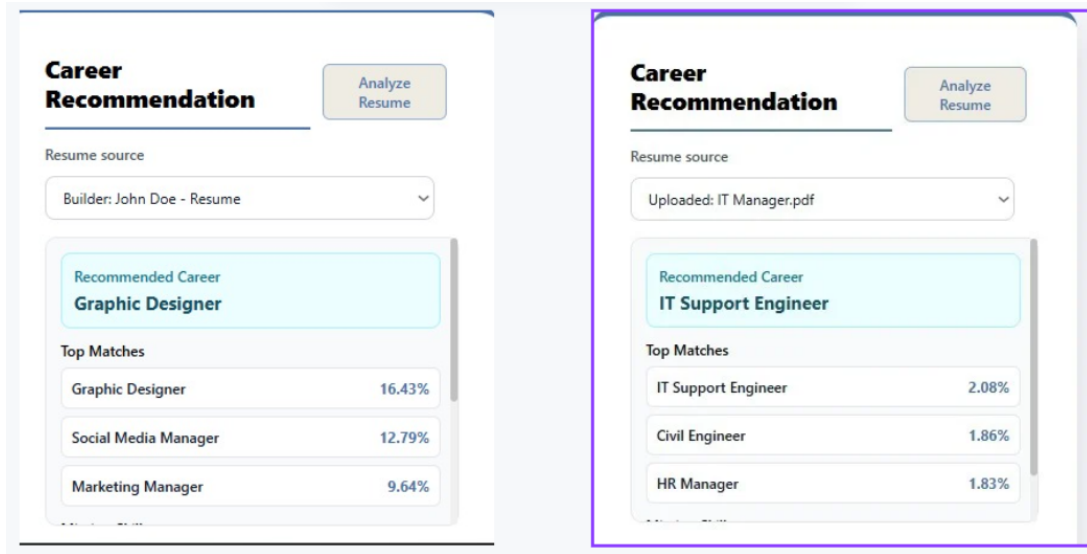


Figure 5.1: Career Recommendation

### 5.1.1 Performance Metrics and Model Evaluation

Table 5.1: Performance Comparison of Resume Classification Models

| Model                      | Accuracy     | Precision   | Recall      | F1 Score    | Advantages                                                                                                                         | Limitations                                                                                          |
|----------------------------|--------------|-------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Naive Bayes                | 78.6%        | 0.74        | 0.75        | 0.76        | Very fast training time and efficient for smaller datasets. Performs well when features are independent.                           | Assumes independence between words, which may not hold true in complex textual data such as resumes. |
| Random Forest              | 85.2%        | 0.83        | 0.82        | 0.84        | Handles nonlinear patterns and reduces overfitting through ensemble learning. Works well with noisy datasets.                      | Higher computational complexity and longer training time compared to simpler models.                 |
| <b>Logistic Regression</b> | <b>94.0%</b> | <b>0.92</b> | <b>0.93</b> | <b>0.93</b> | Works effectively with high-dimensional sparse data generated by TF-IDF features and provides interpretable probabilistic outputs. | Performance may decrease for highly non-linear datasets.                                             |

The models were evaluated using commonly used machine learning performance metrics such as Accuracy, Precision, Recall, and F1-score. These metrics provide a balanced

evaluation of the classification performance by considering both correct predictions and classification errors.

From the results shown in Table 5.1, Logistic Regression achieved the highest performance across all evaluation metrics. With an accuracy of 94% and an F1-score of 0.93, the model demonstrates strong capability in correctly identifying the appropriate career domain for candidate resumes.

Logistic Regression was selected as the final classification model because it performs particularly well with high-dimensional textual data generated by TF-IDF vectorization[4]. The model is computationally efficient, scalable, and provides probabilistic outputs that can be used for ranking multiple career options. These characteristics make it well suited for resume classification tasks in large-scale recruitment platforms.

## 5.2 Semantic Resume–Job Matching

Traditional recruitment systems rely heavily on keyword-based matching between resumes and job descriptions. However, such approaches often fail to recognize semantic relationships between words. For example, terms such as “Software Developer” and “Software Engineer” may represent similar roles but may not match directly using keyword-based methods.

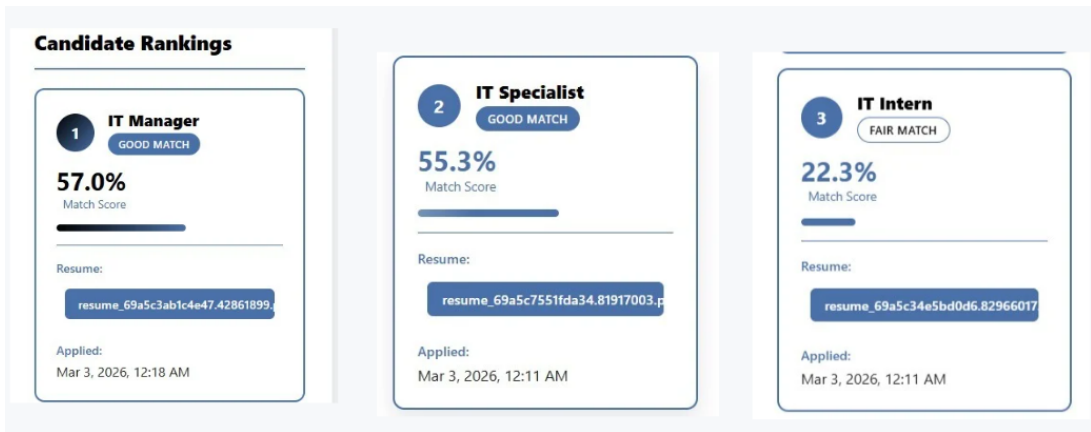


Figure 5.2: Semantic Matching and Ranking

To address this limitation, the JobNexus platform implements semantic matching using the Sentence Transformer model *all-MiniLM-L6-v2*[12]. This transformer-based model converts both resumes and job descriptions into high-dimensional embedding vectors[12,

1]. These vectors capture the contextual meaning of text rather than simply matching individual words, which is later converted to a match score as shown in Figure 5.2.

The similarity between resumes and job descriptions is then calculated using cosine similarity between the embedding vectors[1]. This allows the system to identify conceptually related roles even when different terminology is used.

### 5.2.1 Comparison with Traditional Matching Techniques

The advantages of the semantic matching approach implemented in JobNexus are highlighted in Table 5.2.

Table 5.2: Comparison of Resume Matching Techniques

| Feature               | Keyword Matching                                        | TF-IDF Matching                                    | Semantic Matching (JobNexus)                                          |
|-----------------------|---------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------|
| Matching Principle    | Exact word matching between resume and job description. | Vector similarity using term frequency statistics. | Transformer-based embeddings capturing contextual meaning.            |
| Context Understanding | No contextual understanding.                            | Limited contextual awareness.                      | High semantic understanding of job roles and skills.                  |
| Handling Synonyms     | Cannot detect synonyms or related terms.                | Partially handles similar terms.                   | Effectively recognizes synonyms and related professional terminology. |
| Ranking Method        | Based on keyword occurrence count.                      | Cosine similarity between TF-IDF vectors.          | Cosine similarity between sentence embeddings.                        |
| Matching Accuracy     | Low                                                     | Moderate                                           | High                                                                  |
| System Scalability    | High                                                    | Moderate                                           | High when combined with vector search systems.                        |

The semantic matching approach significantly improves the ability of the system to identify relevant job opportunities by capturing the deeper meaning of resume content and job descriptions.

## 5.3 Skill Gap Analysis and Course Recommendation

Another important feature of the JobNexus platform is the Skill Gap Analysis module. This component compares the skills extracted from a candidate's resume with the skills

required in job descriptions. Natural language processing techniques are used to identify important entities such as technical skills, tools, and programming languages[7, 8].

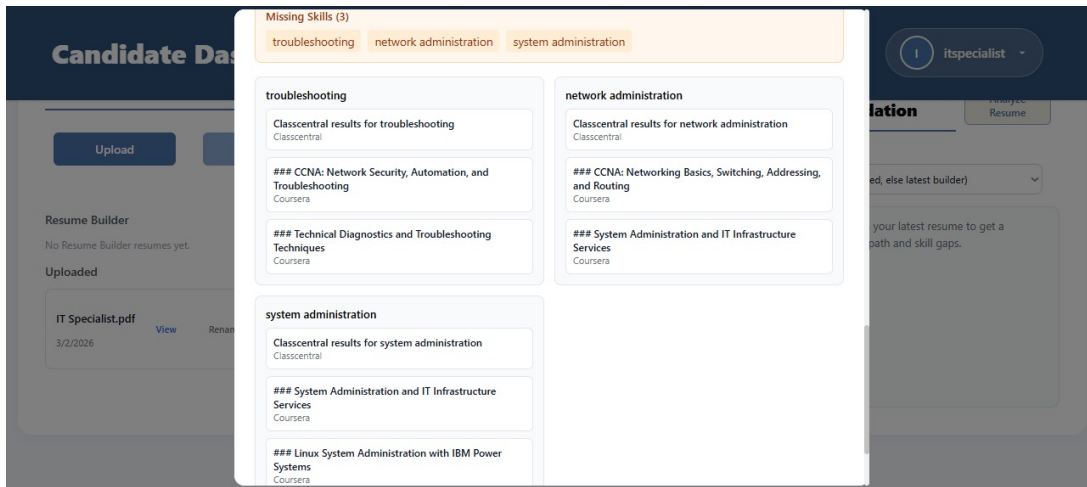


Figure 5.3: Skill Gap Analysis and Course Recommendation

Once missing skills are identified, the system provides recommendations for relevant learning resources[3] as seen in Figure 5.3. The course recommendation module uses FAISS (Facebook AI Similarity Search) to retrieve the most relevant courses based on vector similarity[2].

The comparison of different course retrieval approaches is shown in Table 5.3.

Table 5.3: Comparison of Course Recommendation Methods

| Method                       | Retrieval Speed  | Recommendation Relevance | Scalability      | Remarks                                                                                                       |
|------------------------------|------------------|--------------------------|------------------|---------------------------------------------------------------------------------------------------------------|
| Keyword Search               | High             | Low                      | Moderate         | Depends entirely on exact course titles and may miss conceptually relevant courses.                           |
| TF-IDF Retrieval             | Moderate         | Moderate                 | Moderate         | Captures term importance but does not fully capture semantic meaning.                                         |
| <b>FAISS Semantic Search</b> | <b>Very High</b> | <b>High</b>              | <b>Very High</b> | Efficient vector similarity search allows fast retrieval of conceptually related courses from large datasets. |

This semantic recommendation approach ensures that candidates receive learning resources that are conceptually aligned with their skill gaps rather than simply matching course titles.

### 5.4 Overall System Performance Summary

The overall performance of the JobNexus platform can be summarized by evaluating the individual modules that contribute to the recruitment and career guidance process as shown in Table 5.4.

Table 5.4: Overall System Performance Summary

| Module                | Technique Used                  | Performance Outcome                                                       |
|-----------------------|---------------------------------|---------------------------------------------------------------------------|
| Resume Classification | TF-IDF + Logistic Regression    | Achieved 94% classification accuracy in identifying candidate domains.    |
| Resume-Job Matching   | Sentence Transformer (MiniLM)   | High semantic similarity detection enabling accurate job recommendations. |
| Skill Gap Analysis    | NLP-based skill extraction      | Accurate identification of missing skills from candidate profiles.        |
| Course Recommendation | FAISS Vector Similarity Search  | Highly relevant course suggestions based on identified skill gaps.        |
| Career Recommendation | Multinomial Logistic Regression | Reliable prediction of suitable career paths for candidates.              |

### 5.5 Discussion

The experimental results demonstrate that the integration of machine learning and natural language processing techniques significantly improves the effectiveness of automated recruitment systems. The use of TF-IDF vectorization combined with Logistic Regression enabled accurate classification of resumes into relevant professional categories. Additionally, the implementation of transformer-based semantic matching addressed the common problem of vocabulary mismatch between resumes and job descriptions.

The Skill Gap Analysis and Course Recommendation modules further enhance the value of the system by helping candidates identify areas for improvement and providing actionable learning resources. This dual functionality supports both recruiters and job seekers, making the platform more comprehensive than traditional applicant tracking systems.

## 5.6 Summary

This chapter presented the evaluation of the JobNexus platform and demonstrated how the integration of artificial intelligence improves the recruitment process. The resume classification module, built using TF-IDF and Logistic Regression, achieved an accuracy of 94%, showing its effectiveness in predicting suitable career domains and supporting data-driven recommendations.

The system’s semantic resume–job matching approach, based on sentence transformer embeddings, significantly outperformed traditional keyword-based methods by understanding context and identifying related roles even with different terminology. This improves the accuracy and relevance of job recommendations.

In addition, the skill gap analysis module effectively identified missing skills from candidate profiles, while the course recommendation system, powered by FAISS, provided relevant and targeted learning resources to help candidates improve.

Overall, the results highlight that combining machine learning and NLP creates a more intelligent and efficient recruitment system. JobNexus enhances both hiring decisions for recruiters and career development for candidates by providing accurate matching, meaningful insights, and actionable recommendations.

## Chapter 6

### Conclusions & Future Scope

#### 6.1 Conclusion

The JobNexus platform was developed to address limitations in traditional recruitment systems by integrating artificial intelligence and natural language processing techniques into the job search and candidate evaluation process. The system provides an intelligent recruitment environment where candidates can build and manage resumes, search for job opportunities, analyze skill gaps, prepare for interviews, and receive career recommendations. At the same time, recruiters are provided with tools to create job postings, evaluate applicants, and rank candidates based on AI-generated compatibility scores.

The implementation of transformer-based semantic matching using sentence embeddings enables the system to understand the contextual relationship between resumes and job descriptions rather than relying solely on keyword matching. Additional modules such as skill gap analysis, course recommendation using FAISS-based similarity search, interview preparation, and career prediction extend the system beyond a traditional recruitment portal. The results obtained from testing the platform demonstrate that the system can effectively assist both candidates and recruiters by improving job matching accuracy, identifying missing skills, and providing structured career guidance. Overall, the JobNexus platform demonstrates how AI-driven technologies can enhance modern recruitment systems and support better decision-making in the hiring process.

#### 6.2 Future Scope

Although the current system provides several intelligent features, there are multiple opportunities for further improvement and expansion. Future work could include integrating large language models to generate more advanced interview feedback and personalized career advice. The platform could also incorporate real-time job market analytics and



expand the course recommendation system by connecting to external learning platforms. Additionally, multilingual resume processing and support for additional file formats could improve accessibility for a wider range of users. Implementing more advanced deep learning models and expanding training datasets could further enhance the accuracy of resume matching and career prediction.

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# Appendix A: Presentation



# JOB NEXUS: AI POWERED HIRING AND CAREER ASSISTANCE PLATFORM

## Final Project Evaluation

### Members:

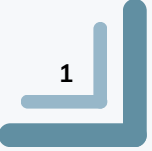
Nandana Anukumar(U2203147)  
Nikhitha Konnikkara Chandran(U2203160)  
Rinu Tony(U2203180)  
Sruthilaya Sunil Kumar(U2203200)

### Guide:

Dr.Mary Priya Sebastian

3rd March, 2026

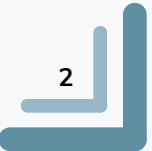
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## CONTENTS

- Problem Definition
- Project Objective
- Purpose and Need
- Literature Survey
- Proposed Method
- Architecture Diagram
- Module description and diagrams
- Results and Outputs
- Assumptions
- Work breakdown and responsibilities
- Hardware and Software Requirements
- Gantt Chart
- Risks and Challenges
- Conclusion
- References

2

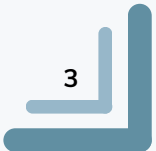




# PROBLEM DEFINITION

Traditional recruitment processes are often time-consuming, inefficient, and difficult to scale, as recruiters must manually review large numbers of resumes to identify suitable candidates. This manual approach can lead to inconsistent evaluations, overlooked talent, and delays in the hiring process. At the same time, job seekers often lack clear feedback on how well their skills match job requirements or what they need to improve. These challenges highlight the need for an intelligent system that can automate resume analysis, accurately match candidates with job descriptions, and provide transparent insights to improve the efficiency and fairness of the recruitment process.

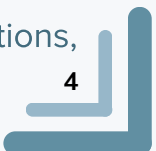
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# OBJECTIVES

1. Develop an AI-powered system to automate resume processing, semantic job description matching, and objective candidate ranking.
2. Provide transparent match scores with reasoning, ensuring fair, explainable, and data-driven recruitment decisions.
3. Enable skill gap identification with personalized course recommendations and structured upskilling guidance.
4. Support complete recruiter workflow including job posting, job description, applicant tracking, notifications, and automated ranking.
5. Assist candidates with AI-based interview preparation, career recommendations, and continuous profile improvement support.

4





# PURPOSE AND NEED

- Overcoming the inefficiencies of traditional hiring processes that rely heavily on manual resume screening.
- Supporting recruiters in objectively evaluating and shortlisting candidates from large volumes of applications.
- Ensuring a transparent and data-driven system for fair candidate evaluation and ranking.
- Automation of resume processing and semantic matching between resumes and job descriptions.
- Generation of explainable match scores that clearly indicate candidate suitability for a role.
- Identification of candidate skill gaps along with personalized course recommendations.
- AI-based interview preparation and career guidance to improve candidate readiness.
- Providing recruiter-side features such as job posting, job description generation, application tracking, and notification management.
- Improving the overall efficiency, fairness, and speed of the recruitment process for both recruiters and candidates.

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# LITERATURE SURVEY

**Title: Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings for Precise Candidate Matching (Bevara et al., 2025)**

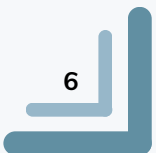
## Advantages

1. Captures semantic meaning of skills and experiences instead of relying on exact keyword matching.
2. Improves precision in candidate ranking by reducing vocabulary mismatch issues.

## Disadvantages

1. Requires high computational resources when using large transformer models (e.g., GPT, LLaMA).

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# LITERATURE SURVEY

| Aspect           | Resume2Vec (Paper)      | JOB NEXUS (Our Implementation)              |
|------------------|-------------------------|---------------------------------------------|
| Matching Type    | Pure semantic embedding | Hybrid semantic + skill-aware scoring       |
| Embedding Models | Large transformers      | Lightweight Sentence-Transformer            |
| Similarity       | Cosine similarity       | Cosine + section weighting + skill coverage |
| Ranking          | Based on similarity     | Calibrated, recruiter-aware ranking         |
| Explainability   | Limited                 | Includes reasoning & gap analysis           |

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# LITERATURE SURVEY

**Title: Towards Efficient Resume Understanding: A Multi-Granularity Multi-Modal Pre-Training Approach (IEEE , 2024)**

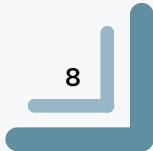
Advantages

- 1. Combines text, layout, and visual features (multi-modal learning), resulting in higher resume parsing accuracy.
- 2. Uses multi-granularity labeling (segment, block, pairwise) for more structured and precise information extraction.

Disadvantages

- 1. Computationally complex due to OCR processing, Faster R-CNN, and Graph Transformer fusion.
- 2. Requires large annotated datasets and high training cost, making real-time deployment challenging.

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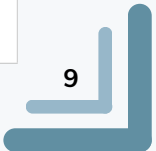




# LITERATURE SURVEY

| Aspect             | Multi-Granularity Multi-Modal Paper (IEEE 2024) | JOB NEXUS (Our Implementation)                       |
|--------------------|-------------------------------------------------|------------------------------------------------------|
| Core Objective     | Automatic structured resume parsing             | Semantic resume-JD matching and ranking              |
| Model Architecture | BERT + Faster R-CNN + Graph Transformer         | Sentence-Transformer (all-MiniLM-L6-v2)              |
| Granularity        | Segment, block, and pairwise labeling           | Document-level + section-level + skill-level scoring |
| Computational Cost | High (multi-modal fusion)                       | Moderate and real-time deployable                    |
| Deployment Focus   | Research-oriented parsing accuracy              | Practical AI recruitment workflow system             |

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# LITERATURE SURVEY

## Title: Application of LLM Agents in Recruitment – A Novel Framework for Automated Resume Screening(2024)

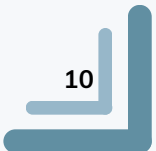
### Advantages

- 1.Can interpret recruiter queries in natural language, making the system flexible and intent-aware.
- 2.Provides transparent ranked recommendations with explanation and continuous refinement using recruiter feedback.

### Disadvantages

- 1.High computational cost and infrastructure requirements due to LLM agent orchestration.
- 2.Dependency on large language models may introduce latency, scalability challenges, and potential hallucination risks.

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# LITERATURE SURVEY

| Aspect             | LLM Agents Recruitment Framework (2024)             | JOB NEXUS (Our Implementation)                                       |
|--------------------|-----------------------------------------------------|----------------------------------------------------------------------|
| Core Approach      | LLM-powered intelligent agents for resume screening | Hybrid semantic matching and calibrated scoring pipeline             |
| Embedding Strategy | Semantic embeddings for resumes and JDs             | Sentence-Transformer (all-MiniLM-L6-v2) embeddings                   |
| Ranking Method     | Similarity-based ranking with LLM reasoning         | Cosine similarity + section weighting + skill coverage + calibration |
| Explainability     | LLM-generated explanations                          | Deterministic reasoning + skill-gap analysis                         |
| Adaptability       | Uses recruiter feedback to refine recommendations   | Uses calibrated scoring and recruiter-required skill boosts          |
| Computational Cost | High (LLM agent orchestration)                      | Moderate and suitable for real-time API deployment                   |

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# LITERATURE SURVEY

**Title: A novel approach for job matching and skill recommendation using transformers and the O\*NET database**

Advantages

- 1. Uses the structured O\*NET occupational taxonomy, enabling standardized and fair skill-based job matching.
- 2. Reduces semantic gaps between resumes and job descriptions by mapping terms across multiple categories (skills, knowledge, abilities, etc.).

Disadvantages

- 1. Heavy dependency on the O\*NET database may limit adaptability to emerging or domain-specific skills.
- 2. Mapping and weighted scoring mechanisms may require complex preprocessing and maintenance for accurate category alignment.

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# LITERATURE SURVEY

| Aspect            | O*NET-Based Job Matching Paper                   | JOB NEXUS (Our Implementation)                                 |
|-------------------|--------------------------------------------------|----------------------------------------------------------------|
| Core Approach     | Uses O*NET taxonomy for structured skill mapping | Uses semantic embeddings with hybrid scoring                   |
| Skill Mapping     | Maps resume terms to predefined O*NET categories | Extracts skills dynamically using NLP + embeddings             |
| Similarity Method | Weighted and normalized category-based scoring   | Cosine similarity + section-level + skill coverage calibration |
| Deployment Focus  | Structured, taxonomy-driven matching             | Real-time, recruiter-aware AI recruitment platform             |

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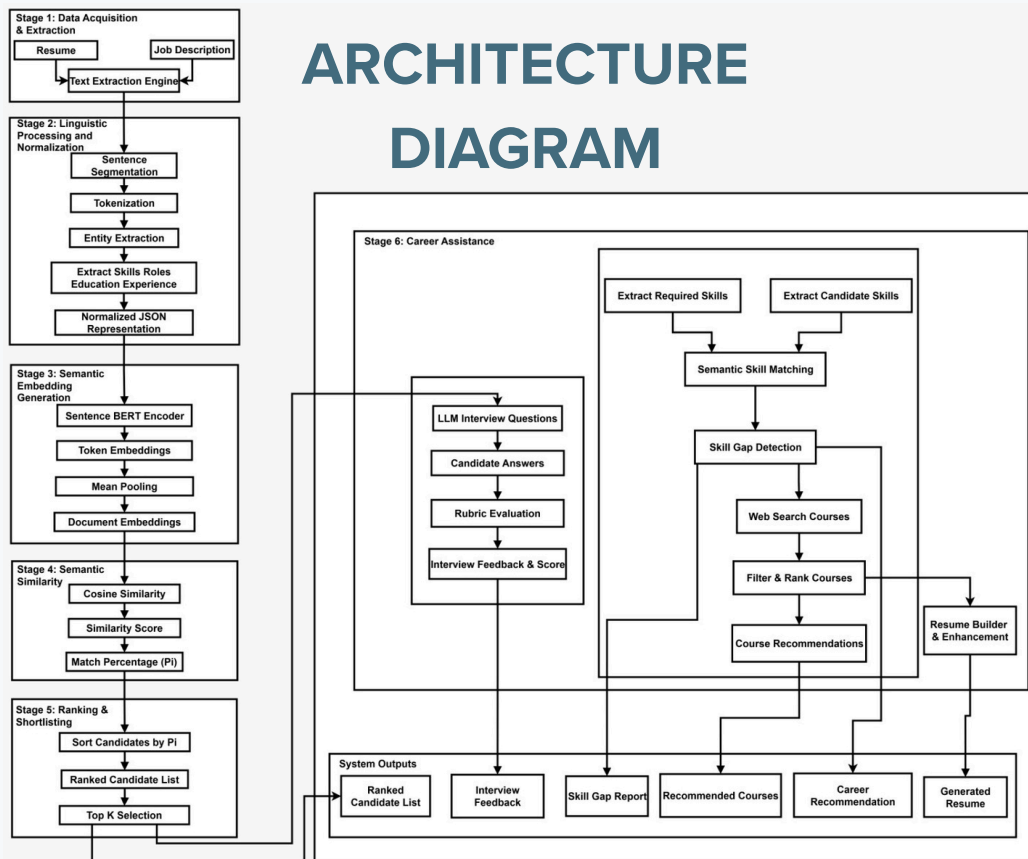


# PROPOSED METHOD

- Development of a modular AI-powered hiring and career assistance platform.
- Integration of a structured **Resume Builder** to generate standardized, ATS-friendly resumes.
- **NLP-based parsing** and data extraction from resumes and job descriptions.
- **Hybrid semantic matching** and similarity scoring to evaluate resume–job alignment.
- Automated **candidate ranking** based on computed **match scores**.
- **Skill-gap analysis** to identify missing or weak skills in candidate profiles.
- Personalized **course recommendations** generated based on identified skill gaps.
- **Career role prediction** using a Logistic Regression model.
- AI-driven **interview preparation**, including role-specific questions and answer evaluation.
- Interactive dashboards displaying match scores, rankings, feedback, and recommendations with secure database support.

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# ARCHITECTURE DIAGRAM



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# MODULE DESCRIPTION

## INPUT MODULE

The Input Module collects and prepares user data for semantic processing and ranking.

- **Candidate Inputs:** Candidates log in and upload resumes (PDF). Files are securely stored before text extraction begins.
- **Recruiter Inputs:** Recruiters provide job details such as job title, description, required skills, and role specifications. The job description is stored in the database for semantic comparison with resumes.

The libraries were evaluated using precision, recall, F1-score, and false positive/negative analysis. pdfplumber provided better structural accuracy and skill detection quality, while spaCy offered a stronger precision-recall balance for skill extraction. Therefore, both were selected for optimal accuracy and deployment efficiency.

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# MODULE DESCRIPTION

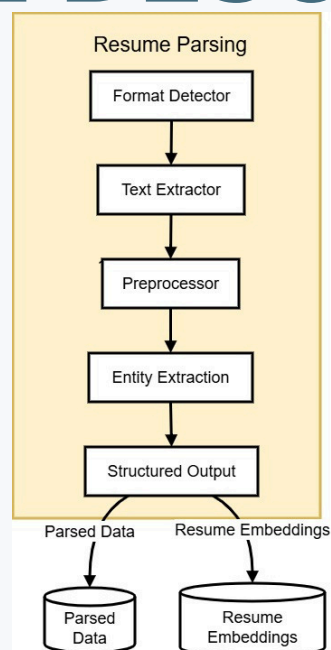
## DATA PARSING AND EXTRACTION PHASE

- After input collection, the module performs structured parsing and information extraction to convert unstructured resume and job description text into machine-readable data.
- Since resumes are typically uploaded as PDF documents, multiple PDF parsing libraries were evaluated during development.

| PyPDF                                                           | PyMuPDF                                                                | Pdfplumber                                                             |
|-----------------------------------------------------------------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| Produced inaccurate text extraction with broken reading order   | Introduced spacing inconsistencies in extracted text                   | Provided more consistent and structured text extraction                |
| Merged multi-column layouts, affecting structure clarity        | Struggled with varied resume layouts and complex formatting            | Handled multi-page resumes more reliably                               |
| Caused formatting loss, leading to inconsistent skill detection | Increased implementation complexity without stable output improvements | Delivered cleaner block reconstruction across diverse resume templates |

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## LINGUISTIC PROCESSING

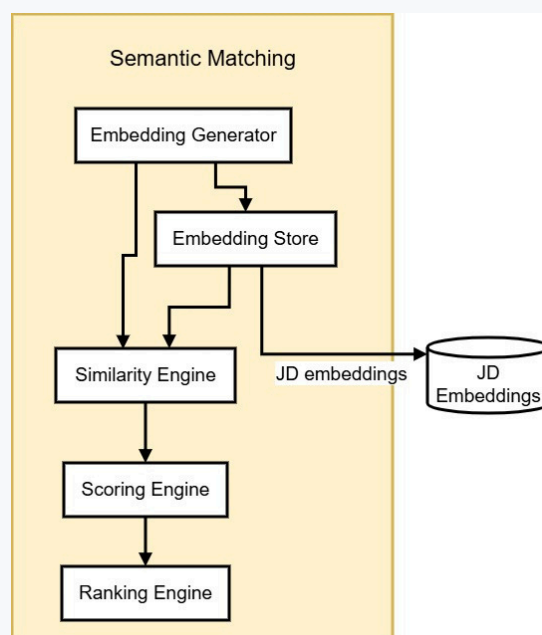
Once raw text is extracted, Natural Language Processing is applied to identify structured information. The system uses:

- **SpaCy** as the primary NLP engine : SpaCy is responsible for extracting candidate skill phrases and relevant technical terms from resumes and job descriptions.
- **NLTK** - Sentence tokenization for clean text segmentation

| NLTK (Rule-Based Approach)                                     | CoreNLP                                                   | spaCy                                                                     |
|----------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------|
| Primarily rule-based tokenization and tagging                  | Java-based NLP framework with strong linguistic accuracy  | Production-ready Python NLP framework                                     |
| Used mainly for sentence tokenization                          | Higher computational overhead and integration complexity  | Fast and efficient for real-time resume processing                        |
| Limited contextual understanding for modern semantic pipelines | Requires external server setup and heavier infrastructure | Supports Noun Chunks, NER, POS tagging, and phrase normalization natively |
| Not ideal for structured skill extraction                      | Overkill for resume-level lightweight extraction tasks    | Used as the primary NLP engine in the system                              |

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## SEMANTIC MATCHING MODULE

Once resume text and job description text are normalized, semantic matching is applied to measure contextual similarity rather than simple keyword overlap.

We evaluated different text representation approaches:

- **TF-IDF (Traditional Vectorization)**
- **BERT (Contextual Transformer Model)**
- **SBERT (Sentence-BERT) – Used in our system**

| Approach                                 | Correlation with Recruiter Score | Precision@5 | Pros                                | Cons                                  |
|------------------------------------------|----------------------------------|-------------|-------------------------------------|---------------------------------------|
| TF-IDF + Cosine                          | 0,61                             | 0,65        | Fast, lightweight                   | Keyword-based, no context             |
| BERT (Base)                              | 0,79                             | 0,84        | Deep contextual understanding       | Heavy, slow                           |
| SBERT (all-MiniLM-L6-v2)                 | 0,82                             | 0,88        | Fast, semantic similarity optimized | Slightly less powerful than full BERT |
| <b>Hybrid (SBERT + Skill Heuristics)</b> | <b>0,85</b>                      | <b>0,90</b> | Best real-world performance         | Slightly complex logic                |

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# MODULE DESCRIPTION

## SKILL GAP ANALYSIS:

After semantic matching is completed, the system analyzes the difference between job-required skills and candidate skills.

The objective is to identify missing or weak skills and provide structured improvement insights.

We evaluated different skill comparison approaches:

- **Exact Keyword Matching**
- **Lexical Similarity (Fuzzy Matching)**
- **Semantic Similarity (Embedding-Based) – Used in our system**

| Exact Matching (Rule-Based)  | Lexical Similarity (Fuzzy)            | Semantic Similarity (Embedding-Based)                |
|------------------------------|---------------------------------------|------------------------------------------------------|
| Checks exact string match    | Measures character similarity         | Measures contextual similarity                       |
| Fast and simple              | Handles minor spelling variations     | Detects conceptual equivalence                       |
| Fails if wording differs     | Still fails for semantic variations   | Identifies related skills (e.g., DL ≈ Deep Learning) |
| No contextual understanding  | Limited contextual awareness          | Context-aware and intelligent                        |
| Suitable for basic filtering | Moderate improvement over exact match | Used in our system for accurate skill-gap detection  |

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# MODULE DESCRIPTION

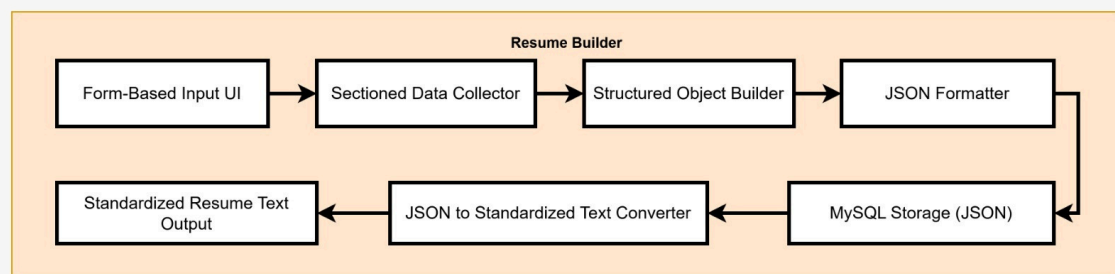
## RESUME BUILDER:

- Designed as a structured, form-based resume creation system instead of free-form PDF uploads.
- Collects information through predefined sections such as personal details, education, experience, skills, and template preferences.
- Stores all resume data in MySQL using JSON format to preserve structure and support nested entries.
- Maintains clear separation of sections for consistency and easy updates.
- Converts structured JSON data into standardized plain text before AI processing.
- Acts as a normalization layer to improve semantic matching accuracy and skill extraction reliability.
- Ensures clean, consistent, and AI-ready resume data across the platform.

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## COURSE RECOMMENDATION MODULE

After identifying missing skills through skill-gap analysis, the system recommends relevant learning resources.

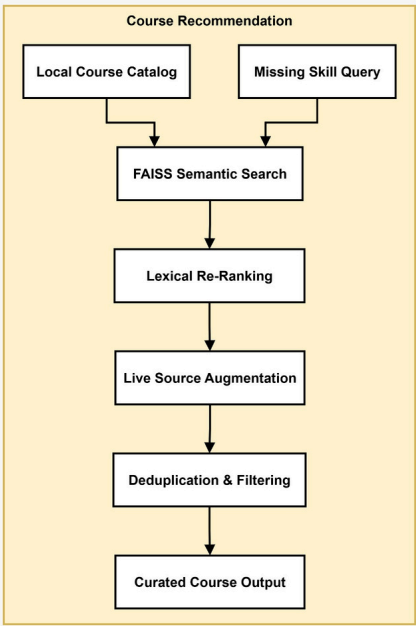
We evaluated different recommendation approaches:

- **Manual Skill-to-Course Mapping**
- **Keyword-Based Search**
- **Vector Search using FAISS (Used in our system)**

| Manual Mapping                              | Keyword-Based Search                  | FAISS (Vector Search)            |
|---------------------------------------------|---------------------------------------|----------------------------------|
| Predefined mapping between skill and course | Matches courses using keyword overlap | Uses embedding similarity search |
| Static and hardcoded                        | Limited contextual understanding      | Context-aware retrieval          |
| Difficult to scale                          | May return irrelevant results         | Fast nearest-neighbor search     |
| Requires manual updates                     | Sensitive to wording variations       | Efficient for large datasets     |
| Not adaptive                                | No semantic intelligence              | Used in our system               |

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## INTERVIEW PREPARATION MODULE

This module generates role-specific interview questions and evaluates candidate responses.

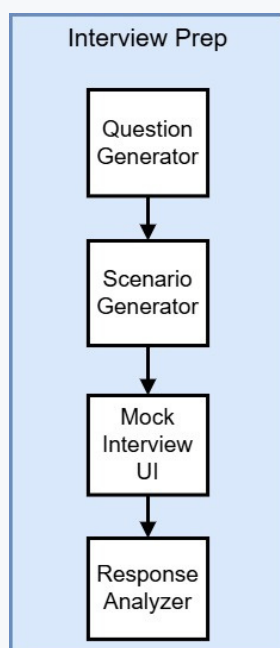
We considered:

- **Static Question Bank**
- **Rule-Based Evaluation**
- **AI-Based Dynamic Generation & Evaluation (Used in our system)**

| Static Question Bank      | Rule-Based Evaluation    | AI-Based Generation (Used)  |
|---------------------------|--------------------------|-----------------------------|
| Fixed set of questions    | Predefined scoring logic | Dynamic question generation |
| Same for all users        | Limited flexibility      | Role-specific questions     |
| No personalization        | No contextual evaluation | Context-aware feedback      |
| Cannot adapt to new roles | Hardcoded logic          | Scalable and intelligent    |
| Limited usefulness        | Basic automation         | Used in our system          |

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## CAREER RECOMMENDATION MODULE

This module predicts suitable career roles based on candidate skills, resume content, and profile features.

We evaluated:

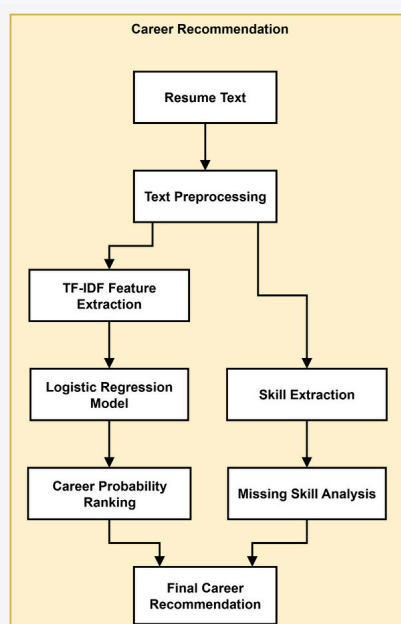
- Rule-Based Career Mapping
- Deep Learning Models
- Logistic Regression (Used in our system)

| Approach                          | Accuracy     | F1-Score     | Pros                                               | Cons                                 |
|-----------------------------------|--------------|--------------|----------------------------------------------------|--------------------------------------|
| Naive Bayes                       | 78.6%        | 0.76         | Fast, simple                                       | Lower accuracy, assumes independence |
| SVM (Linear)                      | 84.2%        | 0.83         | Good with text data                                | Slower training                      |
| Random Forest                     | 82.5%        | 0.81         | Handles non-linearity                              | Less interpretable                   |
| <b>Logistic Regression (Used)</b> | <b>87.4%</b> | <b>0.855</b> | <b>Interpretable, stable, probabilistic output</b> | <b>Linear decision boundary</b>      |

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# MODULE DESCRIPTION



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# MODULE DESCRIPTION

## OUTPUT MODULE

- **Resume Builder Output:** Generates structured, ATS-friendly resumes that can be used for job applications.
- **Match Score and Candidate Ranking :** Provides resume–job match scores and automatically ranks candidates based on semantic similarity and relevance.
- **Skill Gap Analysis Results :** Identifies missing or weak skills in a candidate’s profile compared to job requirements.
- **Personalized Course Recommendations:** Suggests relevant courses and learning resources to help candidates improve identified skill gaps.
- **Career Role Prediction:** Predicts suitable career roles for candidates based on their skills and profile using the trained model.
- **Interview Preparation Support:** Generates role-specific interview questions, preparation tips, and evaluates candidate answers with feedback.
- **Application Tracking and Notifications:** Shows application status, screening results, and sends updates or notifications throughout the recruitment process.

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# RESULT AND OUTPUT

## System Output

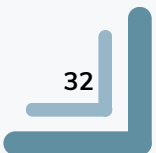
### For Recruiters:

- Ranked list of candidates
- Resume–JD match score with reasoning
- Structured job description support
- Application status tracking

### For Candidates:

- Professional resume creation support
- Skill-gap feedback and improvement suggestions
- Career path recommendations
- Interview preparation materials
- Application progress tracking

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# OUTPUT

## Candidate Dashboard

Find and apply to your dream job

Notifications

itspecialist

### My Resumes

Upload>Create

Resume Builder

No Resume Builder resumes yet.

Uploaded

IT Specialist.pdf

3/2/2026

View

Rename

Delete

### Available Jobs 2

All

Refresh

#### Senior UI Designer

[ ] Unapplied

Deloitte

We are seeking a senior ui designer who relishes in methodically applied color palettes and takes pride in your type of ramp when your work is done the mobile apps and responsive websites you create not only look amazing but act as a seaml...

Show more

### Career Recommendation

Analyze Resume

Resume source

Auto (latest uploaded, else latest builder)

Run AI analysis on your latest resume to get a suggested career path and skill gaps.

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### Candidate Dashboard

Upload>Create

Resume Builder

No Resume Builder resumes yet.

Uploaded

IT Specialist.pdf

3/2/2026

View

Rename

Delete

### Match Score Analysis

Job

57.8%

Match Score

Moderate Match - Some relevant experience

#### Match Analysis

**Score Explanation**

Score 0.58 indicates moderate match - some relevant experience. Local analysis found 1 detected skills and 13 overlapping terms between resume and job description.

### Candidate Dashboard

Upload>Create

Resume Builder

No Resume Builder resumes yet.

Uploaded

IT Specialist.pdf

3/2/2026

View

Rename

Delete

### Missing Skills (3)

troubleshooting

network administration

system administration

#### troubleshooting

Classcentral results for troubleshooting

Classcentral

### CCNA: Network Security, Automation, and Troubleshooting

Coursera

### Technical Diagnostics and Troubleshooting Techniques

Coursera

#### network administration

Classcentral results for network administration

Classcentral

### CCNA: Networking Basics, Switching, Addressing, and Routing

Coursera

### System Administration and IT Infrastructure Services

Coursera

#### system administration

Classcentral results for system administration

Classcentral

### System Administration and IT Infrastructure Services

Coursera

### Linux System Administration with IBM Power Systems

Coursera

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Career Recommendation

Analyze Resume

Resume source

Builder: John Doe - Resume

Recommended Career

Graphic Designer

Top Matches

|                      |        |
|----------------------|--------|
| Graphic Designer     | 16.43% |
| Social Media Manager | 12.79% |
| Marketing Manager    | 9.64%  |

Career Recommendation

Analyze Resume

Resume source

Uploaded: IT Manager.pdf

Recommended Career

IT Support Engineer

Top Matches

|                     |       |
|---------------------|-------|
| IT Support Engineer | 2.08% |
| Civil Engineer      | 1.86% |
| HR Manager          | 1.83% |

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Resume Builder

Create a professional resume step by step

G graphicdesigner2

1 Personal Info

2 Experience

3 Education

4 Skills

5 Template

Personal Information

Enter your basic information. This will appear at the top of your resume.

First Name \*

John

Last Name \*

Doe

Email \*

Phone \*

(123) 456-7890

36

## Resume Builder

Create a professional resume step by step

G

graphicdesigner2

### Experience #1



Company Name \*

Company Name, City, State



Position \*

Graphic Designer (Volunteer)



Start Date \*

April, 2015



End Date

-----, ----



Current  
Position

#### Responsibilities & Achievements

- Designed promotional poster and flyer for a scl

+ Add Another Bullet Point

37

## Resume Builder

Create a professional resume step by step

G

graphicdesigner2



Personal Info



Experience



Education



Skills



Template

### Education

List your educational qualifications in reverse chronological order.

#### Education #1



Institution \*

Forest Park High School, City, State, Unite

38

## Resume Builder

Create a professional resume step by step

G

graphicdesigner2

### Skills

Add your key skills. Include technical, soft, and language skills.

#### Your Skills (0)

No skills added yet. Add some skills below.

#### Add New Skill

Enter a skill (e.g., 'Public S

+ Add

39

## Resume Builder

Create a professional resume step by step

G

graphicdesigner2

#### Popular Skills

Q Search skills...

JavaScript

React

Node.js

Python

Java

SQL

AWS

Docker

Git

HTML/CSS

TypeScript

MongoDB

Communication

Leadership

Problem Solving

Teamwork

Project Management

Agile/Scrum

Data Analysis

Machine Learning

40



## Resume Builder

Create a professional resume step by step

G

graphicdesigner2

### ATS-Friendly

#### ATS-Friendly

Clean structure optimized for applicant tracking systems

Select

### Professional

#### Professional

Classic layout for corporate and business roles

Select

### Creative

#### Creative

Unique design for creative fields

✓ Selected

### Minimalist

#### Minimalist

Minimal styling with high readability

Select

41

## Interview Preparation

Practice answers and review interview sessions

Back to Dashboard

G

graphicdesigner2

### Preparing for: Graphic Designer

#### Interview in Progress

Stop Interview

Question 1 of 10

Overall Score: 0/10

#### Question 1:

Our design work heavily relies on the Adobe Creative Suite, including Photoshop, Illustrator, and InDesign. Could you describe your proficiency with these tools and provide examples of projects where you've utilized them?

#### Your Answer:

Type your answer here

42

## Interview Preparation

Practice answers and review interview sessions

[Back to Dashboard](#)

G graphicdesigner2

### Interview Complete

Final Score: 6.7/10

### Interview Review

Close

Final Score: 6.7/10

**Q: Our design work heavily relies on the Adobe Creative Suite, including Photoshop, Illustrator, and InDesign. Could you describe your proficiency with these tools and provide examples of projects where you've utilized them?**

**Answer:** I have hands-on experience using Adobe Creative Suite tools, specifically Photoshop and Illustrator, through my coursework in Graphic Design and my volunteer design projects. In my Graphics/IT program at Forest Park High School, I used Photoshop for photo editing, image manipulation, color correction, and creating visually appealing compositions. I also used Illustrator to design vector-based graphics, including logos, posters, and layout elements that required clean scaling and precise typography. One of my most meaningful projects was designing a poster and flyer for my school's Walk for Suicide Awareness event. For this project, I used Illustrator to create a clear and impactful layout that communicated event details such as time, location, and purpose, while maintaining a strong visual theme. I carefully selected typography, colors, and imagery to ensure the design appealed to a wide audience. The final design

43

## Interview Preparation

Practice answers and review interview sessions

[Back to Dashboard](#)

G graphicdesigner2

**Q: While your resume highlights your experience, we'd love to see examples of your creative work. Could you walk us through a few pieces from your portfolio, explaining your design process and the impact they achieved?**

**Answer:** 1. Walk for Suicide Awareness Poster For this project, I designed a poster and flyer for a school awareness event. I started by researching the theme and identifying key information that needed to stand out, like the event time and location. I used Illustrator to create a clean layout with strong typography and meaningful color choices. The design was selected and shared on social media, helping promote attendance and awareness for the event. 2. Graphic Design Class Projects In my graphic design coursework, I created weekly projects such as logos and promotional graphics. My process usually began with sketching ideas, then refining them digitally in Photoshop or Illustrator. I focused on balance, color harmony, and visual appeal. Some of my work was featured in school showcases, which reinforced that my designs connected well with viewers. 3. Photography & Image Editing Work I've also worked on photography-based designs where I edited and enhanced images in Photoshop. My process involved adjusting lighting, contrast, and composition to create visually appealing results. These projects improved my attention to detail and strengthened my ability to create designs that attract and hold attention.

#### Feedback:

Score: 7/10

**Strengths:** You effectively describe your design process for each project, from initial research or sketching to digital execution. | You clearly name specific design tools like Illustrator and Photoshop, demonstrating your technical skills. | Your examples cover both print and digital media, including social media promotion, which aligns well with the job description. | You highlight the positive impact of your designs, such as promoting event attendance and receiving recognition in showcases.

**Improvements:** To strengthen your answer, explicitly link your project examples to specific job description requirements, such as designing marketing materials, branding assets, or website graphics. | Expand on how you adhere to or establish brand guidelines in your design process, as this is a significant part of the job description and your resume. | Consider including an example of a project that specifically involves website graphics, as this is mentioned in the job description and would showcase a broader range of your digital design capabilities. | For your photography and image editing work, focus more on the external impact or achievement of the project rather than primarily on your personal skill

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## Interview Preparation

Practice answers and review interview sessions

Back to Dashboard

G graphicdesigner2

**AI Feedback (Score: 7/10)**

**Strengths:**

You clearly articulated your proficiency with Photoshop and Illustrator, providing specific uses like photo editing, vector graphics, and typography.

The detailed example of the 'Walk for Suicide Awareness' poster and flyer effectively demonstrates your application of skills, creative process, and ability to meet deadlines and communicate effectively through design.

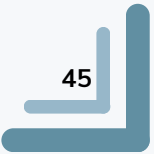
You proactively addressed your familiarity with InDesign, showing an understanding of layout principles and a willingness to adapt, which is a positive trait.

**Areas for Improvement:**

You stated that your resume primarily lists Photoshop and Illustrator, but these tools are not explicitly mentioned in the skills section of your provided resume. It's important to ensure your answer accurately reflects your resume, or to update your resume to include your software proficiency.

The coursework and high school program you mentioned in your answer are not detailed in the education section of your resume. Aligning your answer with the educational and professional experiences listed on your resume would create a more consistent and credible profile.

To strengthen your answer, consider integrating examples from your professional Graphic Designer roles listed on your resume, rather than solely relying on coursework and volunteer projects. This would showcase



## Candidate Dashboard

Find and apply to your dream job

Notifications 6

S salesassociate

Upload

Create

**Resume Builder**

No Resume Builder resumes yet.

**Uploaded**

Sales Associate Resu...

View

Rename

Delete

3/3/2026

**Jobs 3**

**Retail Sales Associate**

Urban Trend

We are seeking a Sales customers, meet sales store operations. Resp operating a cash regist card transactions, man stocking shelves, follo.

Show more

**Notifications** 6 unread

Your application for Retail Sales Associate has been reviewed. Recruiter feedback: We have reviewed your resume, further communications will be done through mail  
3/3/2026, 12:47:24 AM  
Mark as read

A recruiter updated the job you applied for: Retail Sales Associate.  
3/3/2026, 12:46:23 AM  
Mark as read

A recruiter updated the job you applied for: Retail Sales Associate.  
3/3/2026, 12:45:28 AM  
Mark as read

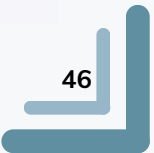
A recruiter updated the job you applied for: Retail

**Foundation**

Resume

loaded, else latest builder)

is on your latest resume to get a eer path and skill gaps.



## Recruiter Dashboard

Manage job postings and review candidate applications

Notifications 4

vitalwerks

### Post New Job

#### Job Title

e.g., Senior Frontend Developer

#### Job Description

Describe the job responsibilities, requirements, and benefits...

Required Skills (comma-separated)

### Applied Candidates

#### FILTERS

##### Job

IT Specialist (v)

##### Status

All

#### IT Specialist

Applications: 4  
candidates

4

CANDIDATES

### Ranking

#### Rank Candidates by Match Score

Select a job and rank candidates who applied for it

##### Job

IT Specialist (Vitalwerks Internet !)

47

## Posted Jobs

Manage all jobs you posted from one page.

Back to Dashboard

vitalwerks

### Senior Software Engineer

Vitalwerks Internet Solutions | 3/2/2026

#### Accepting applications

We are looking for a fulltime full stack senior software engineer who can work out of our office in reno nevada relocation assistance is available you should enjoy writing beautiful fast php to a high standard in a timel...

[Show more](#)

Close Applications

Edit

Delete

48



Candidate Rankings

1

**IT Manager**  
GOOD MATCH

57.0%  
Match Score

Resume:  
resume\_69a5c3ab1c4e47.42861899.p

Applied:  
Mar 3, 2026, 12:18 AM

2

**IT Specialist**  
GOOD MATCH

55.3%  
Match Score

Resume:  
resume\_69a5c7551fda34.81917003.p

Applied:  
Mar 3, 2026, 12:11 AM

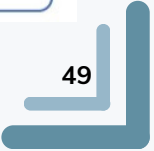
3

**IT Intern**  
FAIR MATCH

22.3%  
Match Score

Resume:  
resume\_69a5c34e5bd0d6.82966017

Applied:  
Mar 3, 2026, 12:11 AM



4

**Teacher T**  
FAIR MATCH

9.2%  
Match Score

Resume:  
resume\_69a5c5a073fb70.51299770.p

Applied:  
Mar 3, 2026, 12:15 AM

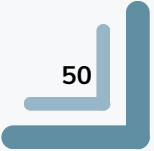
5

**Mechanical Engineer**  
FAIR MATCH

7.1%  
Match Score

Resume:  
resume\_69a5c47179ea19.15104619.p

Applied:  
Mar 3, 2026, 12:14 AM





Database Admin Control Panel

Logout

User Authorization

Database Records

User Authorization Management

Set roles, reset passwords, and delete accounts.

| ID | Email                | Role                 | Name        | Actions                   |
|----|----------------------|----------------------|-------------|---------------------------|
| 26 | urbantrend@gmail.com | <div>recruiter</div> | Recruiter 3 | <div>Reset Password</div> |



Database Admin Control Panel

Logout

User Authorization

Database Records

Database Records Management

jobs

Load Records

Refresh Tables

Primary Key: id

Columns: id, recruiter\_id, title, description, company\_name, created\_at, updated\_at, accepting\_applications, required\_skills

id

recruiter\_id

title

15

26

Retail Sales Associate

id

recruiter\_id

title

14

25

Graphic Designer

id

recruiter\_id

title

13

24

IT Specialist



# ASSUMPTIONS

- Users (candidates, recruiters, and college admin) provide accurate and truthful information.
- Resumes are uploaded in supported formats (PDF) with readable text content.
- Job descriptions clearly specify required skills, qualifications, and experience.
- Semantic similarity scores reasonably reflect candidate suitability.
- AI-generated interview questions and career recommendations are contextually relevant.
- The system database securely stores and manages user data without data loss.
- Users have basic digital literacy to interact with dashboards and features.
- Internet connectivity is available for API-based AI processing.
- The Resume Builder generates ATS-friendly formats compatible with recruitment systems.
- Skill-gap analysis is based only on available resume data (hidden skills cannot be evaluated).

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# WORK DIVISION

- **Nandana Anukumar**

Semantic Matching, Ranking Logic and Evaluation, Career Recommendation Model

- **Nikhitha Konnikkara Chandran**

Resume parsing, Skill and Information extraction, Integration and Testing

- **Rinu Tony**

Resume Builder, Interview prep, Skill gap analysis and Course Recommendations, DB Admin Control Panel, Integration

- **Sruthilaya Sunil Kumar**

Frontend development, UI/UX design, Testing

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# HARDWARE AND SOFTWARE REQUIREMENTS

## HARDWARE:

- Minimum: 8 GB RAM, i5 Processor, 256 GB SSD
- Recommended: 16 GB RAM, i7 Processor, 512 GB SSD
- OS: Windows 10 / Linux / macOS

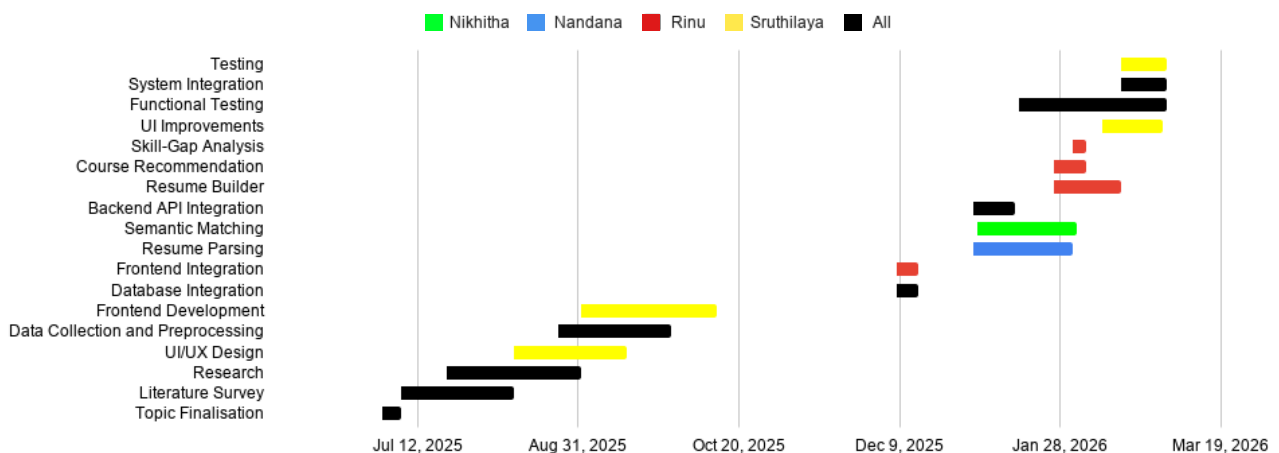
## SOFTWARE AND DATASET:

- Kaggle Dataset
- Python 3.10+
- ReactJS
- Sentence Transformers
- SQL WorkBench
- Git, GitHub, VS Code

55

# GANTT CHART

Gantt Chart - Job Nexus



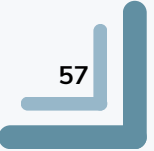
56





# RISKS AND CHALLENGES

- **Bias in AI Models:** AI-based evaluation may unintentionally reflect training data bias.
- **Inaccurate Resume Parsing:** Poor formatting or non-standard resumes may lead to incorrect skill extraction.
- **Semantic Matching Limitations:** Similarity scores still may not fully capture real-world candidate potential.
- **Dependence on External APIs:** AI API downtime or response delays can affect system performance.
- **Scalability Issues:** Handling large volumes of resumes becomes difficult.



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# CONCLUSION

- JOB NEXUS successfully delivers an AI-powered hiring and career assistance platform integrating Resume Builder, Semantic Matching, Similarity Scoring, Candidate Ranking, Skill-Gap Analysis, Course Recommendation, Career Recommendation , and Interview Preparation modules.
- The system streamlines recruitment through intelligent resume analysis, automated ranking, and contextual evaluation.
- It enhances candidate readiness with structured resumes, personalized career predictions, targeted learning recommendations, and role-specific interview support.
- The platform enables faster, fairer, and data-driven hiring decisions.

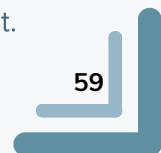


58



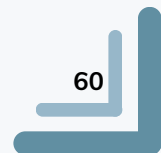
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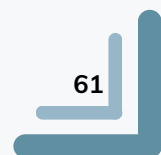
# REFERENCES

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10. van der Meulen, N.; Tona, O.; Leidner, D. E. (2024). Resolving Workforce Skills Gaps with AI-Powered Insights.





**THANK YOU**



## **Appendix B: Vision, Mission, Programme Outcomes and Course Outcomes**

# **Vision, Mission, Programme Outcomes and Course Outcomes**

## **Institute Vision**

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

## **Institute Mission**

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

## **Department Vision**

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

## **Department Mission**

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

## **Programme Outcomes (PO)**

Engineering Graduates will be able to:

**PO1. Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions for complex engineering problems.

**PO2. Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

**PO3. Design/Development of Solutions:** Design creative solutions for complex

engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

**PO4. Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

**PO5. Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling, recognizing their limitations to solve complex engineering problems.

**PO6. The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

**PO7. Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

**PO8. Individual and Collaborative Team Work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

**PO9. Communication:** Communicate effectively and inclusively within the engineering community and society at large; comprehend and write effective reports and design documentation, and make effective presentations considering cultural, language, and learning differences.

**PO10. Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making to one's own work, as a member and leader in a team, and to manage projects in multidisciplinary environments.

**PO11. Life-Long Learning:** Recognize the need for, and have the preparation and ability for (i) independent and life-long learning, (ii) adaptability to new and emerging technologies, and (iii) critical thinking in the broadest context of technological change.

### **Programme Specific Outcomes (PSO)**

A graduate of the Computer Science and Engineering Program will demonstrate:

**PSO1: Computer Science Specific Skills**

The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

**PSO2: Programming and Software Development Skills**

The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

**PSO3: Professional Skills**

The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

**Course Outcomes (CO)**

**Course Outcome 1:** Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

**Course Outcome 2:** Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

**Course Outcome 3:** Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

**Course Outcome 4:** Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

**Course Outcome 5:** Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

**Course Outcome 6:** Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

## Appendix C: CO-PO-PSO Mapping



### CO - PO Mapping

| CO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 | PO 11 |
|----|------|------|------|------|------|------|------|------|------|-------|-------|
| 1  | 2    | 2    | 2    | 1    |      | 2    | 1    | 1    | 1    | 1     | 1     |
| 2  | 2    | 2    | 2    |      | 2    | 3    | 2    | 1    | 1    | 1     | 1     |
| 3  |      |      |      |      |      |      |      | 3    | 2    | 2     | 1     |
| 4  |      | 1    | 2    |      | 3    | 2    | 3    | 2    | 2    | 3     | 1     |
| 5  | 2    | 3    | 3    | 1    | 2    |      |      |      |      | 1     | 3     |
| 6  |      |      | 2    |      |      | 2    | 2    | 2    | 3    | 1     | 1     |

### CO - PSO Mapping

| CO | PSO 1 | PSO 2 | PSO 3 |
|----|-------|-------|-------|
| 1  | 2     | 2     | 2     |
| 2  | 2     | 2     |       |
| 3  |       |       | 3     |
| 4  |       | 2     | 2     |
| 5  | 3     |       |       |
| 6  |       |       | 3     |

### Justification

| Mapping               | Low<br>Medium<br>High | Justification                                                                                                     |
|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| 101003/CS822U.CO1-PO1 | Medium                | Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.    |
| 101003/CS822U.CO1-PO2 | Medium                | Requires identification, formulation and analysis of engineering problems before developing solutions.            |
| 101003/CS822U.CO1-PO3 | Medium                | Involves designing systems or solutions to address identified engineering needs.                                  |
| 101003/CS822U.CO1-PO4 | Low                   | Limited investigation or experimentation may be required to validate the developed models or solutions.           |
| 101003/CS822U.CO1-PO6 | Medium                | Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications. |

| Mapping                | Low<br>Medium<br>High | Justification                                                                                         |
|------------------------|-----------------------|-------------------------------------------------------------------------------------------------------|
| 101003/CS822U.CO1-PO7  | Medium                | Encourages consideration of ethical and professional responsibilities while developing solutions.     |
| 101003/CS822U.CO1-PO8  | Low                   | Some level of teamwork may be required while working on modeling or project tasks.                    |
| 101003/CS822U.CO1-PO9  | Low                   | Requires communication of results and findings through reports or presentations.                      |
| 101003/CS822U.CO1-PO10 | Low                   | Involves basic planning and execution of tasks while implementing solutions.                          |
| 101003/CS822U.CO1-PO11 | Low                   | Encourages independent learning and adapting new knowledge to solve real-world problems.              |
| 101003/CS822U.CO1-PSO1 | Medium                | Uses core computer science principles to analyze problems and design appropriate technical solutions. |
| 101003/CS822U.CO1-PSO2 | Medium                | Involves algorithm design and application of software development practices to implement solutions.   |
| 101003/CS822U.CO1-PSO3 | Medium                | Applies computer science fundamentals in research or innovative problem solving for societal needs.   |
| 101003/CS822U.CO2-PO1  | Medium                | Utilizes engineering knowledge to design and develop products or systems.                             |
| 101003/CS822U.CO2-PO2  | Medium                | Requires analysis of user needs and technological feasibility before development.                     |
| 101003/CS822U.CO2-PO3  | Medium                | Focuses on design and development of solutions addressing real-world requirements.                    |
| 101003/CS822U.CO2-PO5  | Medium                | Requires use of modern engineering and IT tools for system or product development.                    |
| 101003/CS822U.CO2-PO6  | High                  | Emphasizes sustainability and societal impact during solution development.                            |

| Mapping                | Low<br>Medium<br>High | Justification                                                                                            |
|------------------------|-----------------------|----------------------------------------------------------------------------------------------------------|
| 101003/CS822U.CO2-PO7  | Medium                | Ethical and professional responsibilities are considered when developing socially relevant technologies. |
| 101003/CS822U.CO2-PO8  | Low                   | Development activities may require collaboration among team members.                                     |
| 101003/CS822U.CO2-PO9  | Low                   | Involves communicating design ideas and development outcomes.                                            |
| 101003/CS822U.CO2-PO10 | Low                   | Requires basic planning and coordination during development tasks.                                       |
| 101003/CS822U.CO2-PO11 | Low                   | Encourages continuous learning of emerging technologies while building solutions.                        |
| 101003/CS822U.CO2-PSO1 | Medium                | Applies computer science fundamentals to design technically feasible solutions.                          |
| 101003/CS822U.CO2-PSO2 | Medium                | Uses programming and software development practices to implement solutions.                              |
| 101003/CS822U.CO3-PO8  | High                  | Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.            |
| 101003/CS822U.CO3-PO9  | Medium                | Requires effective communication among team members during collaboration.                                |
| 101003/CS822U.CO3-PO10 | Medium                | Team leadership involves project coordination and task management.                                       |
| 101003/CS822U.CO3-PO11 | Low                   | Team interaction supports learning from peers and adapting new approaches.                               |
| 101003/CS822U.CO3-PSO3 | High                  | Professional collaboration and teamwork support innovative computer science solutions.                   |
| 101003/CS822U.CO4-PO2  | Low                   | Requires analysis and planning before executing project tasks.                                           |
| 101003/CS822U.CO4-PO3  | Medium                | Involves designing and implementing project solutions based on planned activities.                       |

| Mapping                | Low<br>Medium<br>High | Justification                                                                               |
|------------------------|-----------------------|---------------------------------------------------------------------------------------------|
| 101003/CS822U.CO4-PO5  | High                  | Requires effective use of modern tools and resources during project execution.              |
| 101003/CS822U.CO4-PO6  | Medium                | Considers societal and environmental impact while implementing solutions.                   |
| 101003/CS822U.CO4-PO7  | High                  | Emphasizes adherence to ethical practices and professional responsibilities.                |
| 101003/CS822U.CO4-PO8  | Medium                | Task execution may involve teamwork and coordination among team members.                    |
| 101003/CS822U.CO4-PO9  | Medium                | Requires communication of progress, outcomes and documentation.                             |
| 101003/CS822U.CO4-PO10 | High                  | Focuses on project management including scheduling, resource allocation and task execution. |
| 101003/CS822U.CO4-PO11 | Low                   | Encourages adaptability and continuous learning during project implementation.              |
| 101003/CS822U.CO4-PSO2 | Medium                | Uses software development practices to execute and manage project tasks.                    |
| 101003/CS822U.CO4-PSO3 | Medium                | Applies computer science knowledge in practical project development environments.           |
| 101003/CS822U.CO5-PO1  | Medium                | Uses engineering knowledge to understand technological limitations and opportunities.       |
| 101003/CS822U.CO5-PO2  | High                  | Requires deep problem analysis and identification of research gaps.                         |
| 101003/CS822U.CO5-PO3  | High                  | Proposes innovative design solutions to address identified gaps.                            |
| 101003/CS822U.CO5-PO4  | Low                   | May involve limited investigation or validation of proposed ideas.                          |
| 101003/CS822U.CO5-PO5  | Medium                | Uses appropriate tools and techniques to analyze and develop innovative solutions.          |
| 101003/CS822U.CO5-PO10 | Low                   | Requires minimal planning while proposing new solution approaches.                          |

| Mapping                | Low<br>Medium<br>High | Justification                                                                           |
|------------------------|-----------------------|-----------------------------------------------------------------------------------------|
| 101003/CS822U.CO5-PO11 | High                  | Strongly promotes independent learning and exploration of emerging technologies.        |
| 101003/CS822U.CO5-PSO1 | High                  | Applies computer science fundamentals to identify and solve complex technical problems. |
| 101003/CS822U.CO6-PO3  | Medium                | Requires presentation of design concepts and developed solutions.                       |
| 101003/CS822U.CO6-PO6  | Medium                | Communication may include discussion of societal or environmental implications.         |
| 101003/CS822U.CO6-PO7  | Medium                | Requires responsible and ethical communication of technical results.                    |
| 101003/CS822U.CO6-PO8  | Medium                | Communication supports collaboration within team environments.                          |
| 101003/CS822U.CO6-PO9  | High                  | Strongly relates to effective written and oral communication of technical findings.     |
| 101003/CS822U.CO6-PO10 | Low                   | Requires structured reporting and documentation in project management contexts.         |
| 101003/CS822U.CO6-PO11 | Low                   | Encourages continuous improvement in communication and presentation skills.             |
| 101003/CS822U.CO6-PSO3 | High                  | Communication of innovative computer science solutions and research outcomes.           |